

An introduction to Cryptography



from substitutions to public keys

An introduction to Cryptography

Daniel Hutmacher

An iceberg floating in a calm, blue ocean under a clear sky. The visible tip of the iceberg is on the right side of the frame. A speech bubble points to a small, jagged peak on the left side of this visible tip, which is enclosed in a dashed white box. The vast, hidden portion of the iceberg extends deep into the water, appearing much larger and more complex than the visible part. Sunlight rays penetrate the water from the left, illuminating the submerged structure.

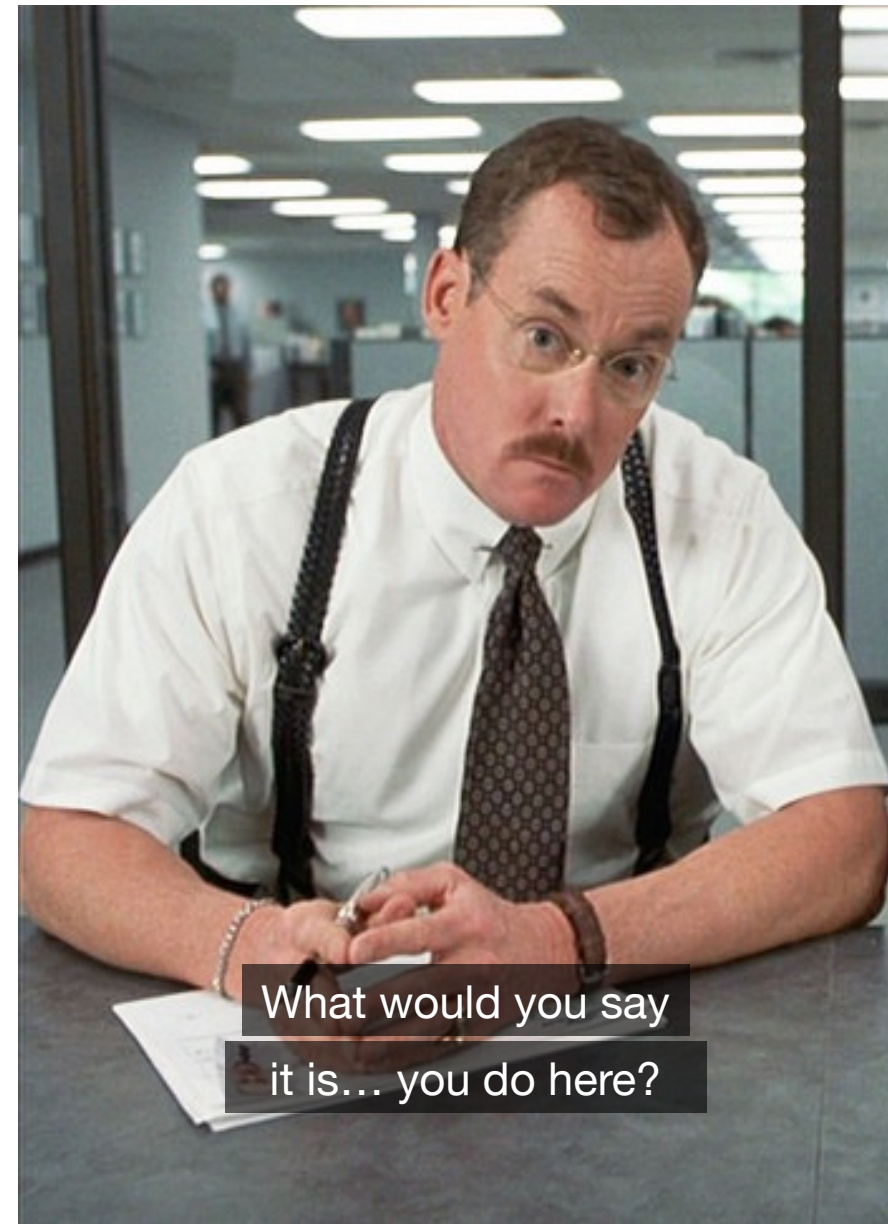
This presentation

Cryptography is pretty complex

Daniel Hutmacher

- SQL Server developer since 1999
- Actually a Data Platform MVP
- Consultant in my own business
- Organizer of Data Saturday Stockholm
- I run datapatform.social and callfordataspeakers.com

Email: daniel@strd.co
Blog: sqlsunday.com
Bluesky: [@dhma.ch](https://dhma.ch)



Thank you, sponsors!

Platinum



HEDDA.IO

Gold



SOLISYON

melissa

DATEN-WG

Caesar shift

Plaintext:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a

Key:

Plaintext:

w	e		a	t	t	a	c	k		a	t		d	a	w	n
x	f		b	u	u	b	d	l		b	u		e	b	x	o

Ciphertext:

Substitution

Plaintext:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
y	r	q	p	o	t	i	h	x	e	z	j	d	c	b	w	n	m	v	l	k	u	s	g	f	a

Key:

Plaintext:

w	e		a	t	t	a	c	k		a	t		d	a	w	n
s	o		y	l	l	y	q	z		y	l		p	y	s	c

Ciphertext:

Substitution ► Frequency analysis

Data source: https://en.wikipedia.org/wiki/Letter_frequency

	English	French	German	Spanish	Portuguese	Italian	Turkish	Swedish	Polish	Dutch	Danish	Icelandic	Finnish	Czech	Hungarian
e	12,7%	14,7%	16,4%	12,2%	12,6%	11,8%	8,9%	10,1%	7,9%	18,9%	15,5%	6,4%	8,0%	7,6%	11,6%
a	8,2%	7,6%	6,5%	11,5%	14,6%	11,7%	11,9%	9,4%	9,0%	7,5%	6,0%	10,1%	12,2%	8,4%	8,9%
n	6,7%	7,1%	9,8%	6,7%	4,4%	6,9%	7,5%	8,5%	5,6%	10,0%	7,2%	7,7%	8,8%	6,5%	6,8%
i	7,0%	7,5%	6,6%	6,2%	6,2%	10,1%	8,6%	5,8%	8,3%	6,5%	6,0%	7,6%	10,8%	6,1%	4,3%
r	6,0%	6,7%	7,0%	6,9%	6,5%	6,4%	6,7%	8,4%	4,6%	6,4%	9,0%	8,6%	2,9%	4,8%	2,7%
t	9,1%	7,2%	6,2%	4,6%	4,3%	5,6%	3,3%	7,7%	4,0%	6,8%	6,9%	5,0%	8,8%	5,7%	7,0%
s	6,3%	7,9%	7,3%	8,0%	6,8%	5,0%	3,0%	6,6%	4,3%	3,7%	5,8%	5,6%	7,9%	5,2%	7,0%
o	7,5%	5,8%	2,6%	8,7%	9,7%	9,8%	2,5%	4,5%	7,6%	6,1%	4,6%	2,2%	5,6%	6,7%	3,7%
l	4,0%	5,5%	3,4%	5,0%	2,8%	6,5%	5,9%	5,3%	2,1%	3,6%	5,2%	4,5%	5,8%	3,8%	6,7%
d	4,3%	3,7%	5,1%	5,0%	5,0%	3,7%	4,7%	4,7%	3,3%	5,9%	5,9%	1,6%	1,0%	3,5%	1,9%
m	2,4%	3,0%	2,5%	3,2%	4,7%	2,5%	3,8%	3,5%	2,9%	2,2%	3,2%	4,0%	3,2%	2,4%	3,8%
u	2,8%	6,3%	4,2%	3,9%	3,6%	2,8%	3,2%	1,9%	2,3%	2,0%	2,0%	4,6%	5,0%	2,2%	0,4%
k	1,8%	0,1%	1,4%	0,0%	0,0%	0,0%	4,7%	3,1%	3,4%	2,3%	3,4%	3,3%	5,0%	2,9%	4,9%
g	2,0%	0,9%	3,0%	1,8%	1,3%	1,6%	1,3%	2,9%	1,4%	3,4%	4,1%	4,2%	0,4%	0,1%	3,8%
c	2,8%	3,3%	2,7%	4,0%	3,9%	4,5%	1,0%	1,5%	4,0%	1,2%	0,6%		0,3%	0,7%	0,6%
h	6,1%	0,9%	4,6%	2,0%	1,3%	0,1%	1,2%	2,1%	1,1%	2,4%	1,6%	1,9%	1,9%	1,4%	1,3%
v	1,0%	1,8%	0,8%	1,1%	1,6%	2,1%	1,0%	2,4%	0,0%	2,9%	2,3%	2,4%	2,3%	5,3%	2,3%
p	1,9%	2,5%	0,7%	2,5%	2,5%	3,1%	0,9%	1,8%	3,1%	1,6%	1,8%	0,8%	1,8%	1,9%	0,5%
b	1,5%	0,9%	1,9%	2,2%	1,0%	0,9%	2,8%	1,5%	1,5%	1,6%	2,0%	1,0%	0,3%	0,8%	1,9%
y	2,0%	0,7%	0,0%	1,4%	0,0%	0,0%	3,3%	0,7%	3,9%	0,0%	0,7%	0,9%	1,7%	1,0%	2,6%
z	0,1%	0,3%	1,1%	0,5%	0,5%	1,2%	1,5%	0,1%	5,6%	1,4%	0,0%		0,1%	1,6%	4,3%
f	2,2%	1,1%	1,7%	0,7%	1,0%	1,2%	0,5%	2,0%	0,3%	0,8%	2,4%	3,0%	0,2%	0,1%	0,5%
j	0,3%	0,8%	0,3%	2,5%	0,9%	0,0%	0,0%	0,6%	2,3%	1,5%	0,7%	1,1%	2,0%	1,4%	1,5%
w	2,4%	0,0%	1,9%	0,0%	0,0%	0,0%		0,1%	4,5%	1,5%	0,1%		0,1%	0,0%	
é		1,5%		0,4%	0,3%							0,6%		0,6%	4,3%
á				0,5%	0,1%							1,8%		0,9%	3,4%
ä			0,6%					1,8%					3,6%		
í							5,1%								
ü				0,7%	0,1%							1,6%		1,6%	0,5%

Frequency analysis

	Ciphertext		English
a	0,5%	a	8,2%
b	1,3%	b	1,5%
c		c	2,8%
d	2,0%	d	4,3%
e	0,9%	e	12,7%
f	2,4%	f	2,2%
g	9,0%	g	2,0%
h	6,2%	h	6,1%
i	4,2%	i	7,0%
j		j	0,3%
k	1,4%	k	1,8%
l	8,0%	l	4,0%
m	6,5%	m	2,4%
n	2,0%	n	6,7%
o	3,1%	o	7,5%
p	1,1%	p	1,9%
q	0,8%	q	
r	7,1%	r	6,0%
s	5,9%	s	6,3%
t	1,8%	t	9,1%
u	1,9%	u	2,8%
v	12,2%	v	1,0%
w	4,0%	w	2,4%
x	2,1%	x	

dv'iv ml hgizmtvih gl olev
 blf pml d gsv ifovh zmw hl wl r
 z ufoo xlnnrgnvmg'h dszg r'n gsrmpmt lu
 blf dl fowm'g tv gsrh uiln zmb lgs vi tfb
 r qfhg dzmmz gvoo blf sld r'n uvvormt
 tlggz nzpv blf fmwvihgzmw
 mvevi tlmmz trev blf fk
 mvevi tlmmz ov g blf wldm
 mvevi tlmmz ifm zilfmw zmw wvhvig blf
 mvevi tlmmz nzpv blf xib
 mvevi tlmmz hzb tllwybv
 mvevi tlmmz gvoo z orv zmw sfig blf
 dv'ev pml dm vzxs lgs vi uli hl olmt
 blfi svzig'h yvvm zxs rmt, yfg blf'iv gl hsb gl hzb rg
 rmhrwv, dv ylg s pml d dszg'h yvvm tlrmt lm
 dv pml d gsv tznv zmw dv'iv tlmmz kozb rg
 zmw ru blf zhp nv sld r'n uvvormt
 wlm'g gvoo nv blf'iv gl yormw gl hvv
 mvevi tlmmz trev blf fk
 mvevi tlmmz ov g blf wldm
 mvevi tlmmz ifm zilfmw zmw wvhvig blf
 mvevi tlmmz nzpv blf xib
 mvevi tlmmz hzb tllwybv
 mvevi tlmmz gvoo z orv zmw sfig blf
 mvevi tlmmz trev blf fk
 mvevi tlmmz ov g blf wldm
 mvevi tlmmz ifm zilfmw zmw wvhvig blf
 mvevi tlmmz nzpv blf xib
 mvevi tlmmz hzb tllwybv
 mvevi tlmmz gvoo z orv zmw sfig blf
 dv'ev pml dm vzxs lgs vi uli hl olmt
 blfi svzig'h yvvm zxs rmt, yfg blf'iv gl hsb gl hzb rg
 rmhrwv, dv ylg s pml d dszg'h yvvm tlrmt lm
 dv pml d gsv tznv zmw dv'iv tlmmz kozb rg
 r qfhg dzmmz gvoo blf sld r'n uvvormt
 tlggz nzpv blf fmwvihgzmw

Frequency analysis

	Ciphertext		English
v	12,2%	e	12,7%
g	9,0%	t	9,1%
l	8,0%	a	8,2%
z	7,9%	o	7,5%
r	7,1%	i	7,0%
m	6,5%	n	6,7%
h	6,2%	s	6,3%
s	5,9%	h	6,1%
i	4,2%	r	6,0%
w	4,0%	d	4,3%
o	3,1%	l	4,0%
f	2,4%	c	2,8%
x	2,1%	u	2,8%
n	2,0%	m	2,4%
d	2,0%	w	2,4%
u	1,9%	f	2,2%
t	1,8%	g	2,0%
k	1,4%	y	2,0%
b	1,3%	p	1,9%
p	1,1%	k	1,8%
y	1,0%	b	1,5%
e	0,9%	v	1,0%
q	0,8%	j	0,3%
a	0,5%	z	0,1%

dv'iv ml hgizmtvih gl olev
 blf pml d gsv ifovh zmw hl wl r
 z ufoo xlnnrgnvmg'h dszg r'n gsrmpmt lu
 blf dl fowm'g tv gsrh uiln zmb lgs vi tfb
 r qfhg dzmmz gvoo blf sld r'n uvvormt
 tlggz nzpv blf fmwvihgzmw
 mvevi tlmmz trev blf fk
 mvevi tlmmz ov g blf wldm
 mvevi tlmmz ifm zilfmw zmw wvhvig blf
 mvevi tlmmz nzpv blf xib
 mvevi tlmmz hzb tllwybv
 mvevi tlmmz gvoo z orv zmw sfig blf
 dv'ev pml dm vzxs lgs vi uli hl olmt
 blfi svzig'h yvvm zxs rmt, yfg blf'iv gl l hsb gl hzb rg
 rmhrwv, dv y lgs pml d dszg'h yvvm tlrmt lm
 dv pml d gsv tznv zmw dv'iv tlmmz kozb rg
 zmw ru blf zhp nv sld r'n uvvormt
 wlm'g gvoo nv blf'iv gl l yormw gl hvv
 mvevi tlmmz trev blf fk
 mvevi tlmmz ov g blf wldm
 mvevi tlmmz ifm zilfmw zmw wvhvig blf
 mvevi tlmmz nzpv blf xib
 mvevi tlmmz hzb tllwybv
 mvevi tlmmz gvoo z orv zmw sfig blf
 mvevi tlmmz trev blf fk
 mvevi tlmmz ov g blf wldm
 mvevi tlmmz ifm zilfmw zmw wvhvig blf
 mvevi tlmmz nzpv blf xib
 mvevi tlmmz hzb tllwybv
 mvevi tlmmz gvoo z orv zmw sfig blf
 dv'ev pml dm vzxs lgs vi uli hl olmt
 blfi svzig'h yvvm zxs rmt, yfg blf'iv gl l hsb gl hzb rg
 rmhrwv, dv y lgs pml d dszg'h yvvm tlrmt lm
 dv pml d gsv tznv zmw dv'iv tlmmz kozb rg
 r qfhg dzmmz gvoo blf sld r'n uvvormt
 tlggz nzpv blf fmwvihgzmw

Frequency analysis

Ciphertext		English	
v	12,2%	e	12,7%
g	9,0%	t	9,1%
l	8,0%	a	8,2%
z	7,9%	o	7,5%
r	7,1%	i	7,0%
m	6,5%	n	6,7%
h	6,2%	s	6,3%
s	5,9%	h	6,1%
i	4,2%	r	6,0%
w	4,0%	d	4,3%
o	3,1%	l	4,0%
f	2,4%	c	2,8%
x	2,1%	u	2,8%
n	2,0%	m	2,4%
d	2,0%	w	2,4%
u	1,9%	f	2,2%
t	1,8%	g	2,0%
k	1,4%	y	2,0%
b	1,3%	p	1,9%
p	1,1%	k	1,8%
y	1,0%	b	1,5%
e	0,9%	v	1,0%
q	0,8%	j	0,3%
a	0,5%	z	0,1%

we're na strongers ta lave
 pac know the rcles ond sa da i
 o fcll uammitment's whot i'm thinking af
 pac wacldn't get this fram onp ather gcp
 i jcst wonno tell pac haw i'm feeling
 gatto moke pac cnderstond
 never ganno give pac cy
 never ganno let pac dawn
 never ganno rcn oracnd ond desert pac
 never ganno moke pac urp
 never ganno sop gaadbpe
 never ganno tell o lie ond hcrt pac
 we've knawn eouh ather far sa lang
 pacr heort's been ouhing, bct pac're taa shp ta sop it
 inside, we bath know whot's been gaing an
 we know the gome ond we're ganno ylop it
 ond if pac osk me haw i'm feeling
 dan't tell me pac're taa blind ta see
 never ganno give pac cy
 never ganno let pac dawn
 never ganno rcn oracnd ond desert pac
 never ganno moke pac urp
 never ganno sop gaadbpe
 never ganno tell o lie ond hcrt pac
 never ganno give pac cy
 never ganno let pac dawn
 never ganno rcn oracnd ond desert pac
 never ganno moke pac urp
 never ganno sop gaadbpe
 never ganno tell o lie ond hcrt pac
 we've knawn eouh ather far sa lang
 pacr heort's been ouhing, bct pac're taa shp ta sop it
 inside, we bath know whot's been gaing an
 we know the gome ond we're ganno ylop it
 i jcst wonno tell pac haw i'm feeling
 gatto moke pac cnderstond

Frequency analysis

	Ciphertext		English
v	12,2%	e	12,7%
g	9,0%	t	9,1%
l	8,0%	a	8,2%
z	7,9%	o	7,5%
r	7,1%	i	7,0%
m	6,5%	n	6,7%
h	6,2%	s	6,3%
s	5,9%	h	6,1%
i	4,2%	r	6,0%
w	4,0%	d	4,3%
o	3,1%	l	4,0%
f	2,4%	c	2,8%
x	2,1%	u	2,8%
n	2,0%	m	2,4%
d	2,0%	w	2,4%
u	1,9%	f	2,2%
t	1,8%	g	2,0%
k	1,4%	y	2,0%
b	1,3%	p	1,9%
p	1,1%	k	1,8%
y	1,0%	b	1,5%
e	0,9%	v	1,0%
q	0,8%	j	0,3%
a	0,5%	z	0,1%

we're no strangers to love
 poc know the rcles and so do i
 a fc1l uommitment's what i'm thinking of
 poc wocldn't get this from anp other gcp
 i jcst wanna tell poc how i'm feeling
 gotta make poc cnderstand
 never gonna give poc cy
 never gonna let poc down
 never gonna rcn arocnd and desert poc
 never gonna make poc urp
 never gonna sap goodbpe
 never gonna tell a lie and hcrt poc
 we've known eauh other for so long
 pocr heart's been auhing, bct poc're too shp to sap it
 inside, we both know what's been going on
 we know the game and we're gonna ylap it
 and if poc ask me how i'm feeling
 don't tell me poc're too blind to see
 never gonna give poc cy
 never gonna let poc down
 never gonna rcn arocnd and desert poc
 never gonna make poc urp
 never gonna sap goodbpe
 never gonna tell a lie and hcrt poc
 never gonna give poc cy
 never gonna let poc down
 never gonna rcn arocnd and desert poc
 never gonna make poc urp
 never gonna sap goodbpe
 never gonna tell a lie and hcrt poc
 we've known eauh other for so long
 pocr heart's been auhing, bct poc're too shp to sap it
 inside, we both know what's been going on
 we know the game and we're gonna ylap it
 i jcst wanna tell poc how i'm feeling
 gotta make poc cnderstand

Frequency analysis

Ciphertext		English	
v	12,2%	e	12,7%
g	9,0%	t	9,1%
l	8,0%	a	8,2%
z	7,9%	o	7,5%
r	7,1%	i	7,0%
m	6,5%	n	6,7%
h	6,2%	s	6,3%
s	5,9%	h	6,1%
i	4,2%	r	6,0%
w	4,0%	d	4,3%
o	3,1%	l	4,0%
f	2,4%	c	2,8%
x	2,1%	u	2,8%
n	2,0%	m	2,4%
d	2,0%	w	2,4%
u	1,9%	f	2,2%
t	1,8%	g	2,0%
k	1,4%	y	2,0%
b	1,3%	p	1,9%
p	1,1%	k	1,8%
y	1,0%	b	1,5%
e	0,9%	v	1,0%
q	0,8%	j	0,3%
a	0,5%	z	0,1%

we're no strangers to love
 pou know the rules and so do i
 a full commitment's what i'm thinking of
 pou wouldn't get this from anp other gup
 i just wanna tell pou how i'm feeling
 gotta make pou understand
 never gonna give pou uy
 never gonna let pou down
 never gonna run around and desert pou
 never gonna make pou crp
 never gonna sap goodbpe
 never gonna tell a lie and hurt pou
 we've known each other for so long
 pour heart's been aching, but pou're too shp to sap it
 inside, we both know what's been going on
 we know the game and we're gonna ylap it
 and if pou ask me how i'm feeling
 don't tell me pou're too blind to see
 never gonna give pou uy
 never gonna let pou down
 never gonna run around and desert pou
 never gonna make pou crp
 never gonna sap goodbpe
 never gonna tell a lie and hurt pou
 never gonna give pou uy
 never gonna let pou down
 never gonna run around and desert pou
 never gonna make pou crp
 never gonna sap goodbpe
 never gonna tell a lie and hurt pou
 we've known each other for so long
 pour heart's been aching, but pou're too shp to sap it
 inside, we both know what's been going on
 we know the game and we're gonna ylap it
 i just wanna tell pou how i'm feeling
 gotta make pou understand

Frequency analysis

	Ciphertext		English
v	12,2%	e	12,7%
g	9,0%	t	9,1%
l	8,0%	a	8,2%
z	7,9%	o	7,5%
r	7,1%	i	7,0%
m	6,5%	n	6,7%
h	6,2%	s	6,3%
s	5,9%	h	6,1%
i	4,2%	r	6,0%
w	4,0%	d	4,3%
o	3,1%	l	4,0%
f	2,4%	c	2,8%
x	2,1%	u	2,8%
n	2,0%	m	2,4%
d	2,0%	w	2,4%
u	1,9%	f	2,2%
t	1,8%	g	2,0%
k	1,4%	y	2,0%
b	1,3%	p	1,9%
p	1,1%	k	1,8%
y	1,0%	b	1,5%
e	0,9%	v	1,0%
q	0,8%	j	0,3%
a	0,5%	z	0,1%

we're no strangers to love
 you know the rules and so do i
 a full commitment's what i'm thinking of
 you wouldn't get this from any other guy
 i just wanna tell you how i'm feeling
 gotta make you understand
 never gonna give you up
 never gonna let you down
 never gonna run around and desert you
 never gonna make you cry
 never gonna say goodbye
 never gonna tell a lie and hurt you
 we've known each other for so long
 your heart's been aching, but you're too shy to say it
 inside, we both know what's been going on
 we know the game and we're gonna play it
 and if you ask me how i'm feeling
 don't tell me you're too blind to see
 never gonna give you up
 never gonna let you down
 never gonna run around and desert you
 never gonna make you cry
 never gonna say goodbye
 never gonna tell a lie and hurt you
 never gonna give you up
 never gonna let you down
 never gonna run around and desert you
 never gonna make you cry
 never gonna say goodbye
 never gonna tell a lie and hurt you
 we've known each other for so long
 your heart's been aching, but you're too shy to say it
 inside, we both know what's been going on
 we know the game and we're gonna play it
 i just wanna tell you how i'm feeling
 gotta make you understand

Frequency analysis

Takeaway:

- The more predictable the cleartext is, the easier it is to crack.
- Frequency analysis requires a lot of text to make sense.
- This stuff was cracked 1200 years ago. Don't use it today.





Zimmermann's Gamble

Zimmermann's gamble



Arthur Zimmermann, German foreign secretary in 1916.

- For most of the first world war, the US tried to stay out of the conflict
- Following the sinking of the Lusitania in 1915, Germany pledged to only use their submarines on the surface.

Zimmermann's gamble



Arthur Zimmermann, German foreign secretary in 1916.

- In 1917, though, Germany thought an unrestricted submarine warfare could be decisive in winning the war.

But they really did not want to pick a fight with the US.
So secretary Zimmermann hatched a plan.

Zimmermann's gamble

CLASS OF SERVICE DESIRED		WESTERN UNION TELEGRAM		NEWCOMB CARLTON, PRESIDENT	
Fast Day Message	☒	CHARGE 5389 		RECEIVED No. 5389 CHECK 5389	
Day Letter	☐			Time Filed	
Night Message	☐				
Night Letter	☐				

Patrons should mark an X opposite the class of service desired; OTHERWISE THE TELEGRAM WILL BE TRANSMITTED AS A FAST DAY MESSAGE.

Send the following telegram, subject to the terms on back hereof, which are hereby agreed to

via Galveston

JAN 19 1917

863.2019/524

GERMAN LEGATION
MEXICO CITY

130	13042	13401	8501	115	3528	416	17214	6491	11310
18147	18222	21560	10247	11518	23677	13805	3494	14936	
98092	5905	11311	10392	10371	0302	21290	5161	39695	
23571	17504	11269	18276	18101	0317	0228	17694	4473	
22284	22200	19452	21589	67893	5569	13918	8958	12137	
1333	4725	4458	5905	17166	13851	4458	17149	14471	6706
13850	12224	6929	14991	7382	15857	67893	14218	36477	
5870	17553	67893	5870	5454	16102	15217	22801	17138	
21001	17388	7446	23638	18222	6719	14331	15021	23845	
3158	23552	22096	21604	4797	9497	22464	20855	4377	
23610	18140	22260	5905	13347	20420	39689	13732	20667	
6929	5275	18507	52262	1340	22049	13339	11265	22295	
10439	14814	4178	6992	8784	7632	7357	6926	52262	11267
21100	21272	9346	9559	22464	15874	18502	18500	15857	
2188	5376	7381	98092	16127	13486	9350	9220	76036	14219
5144	2831	17920	11347	17142	11264	7667	7762	15099	9110
10482	97556	3569	3670						

BEPNSTOPFF.

Charge German Embassy.

Zimmermann's gamble

RECEIVED
October 1-8-58
Washington, State Dept.

TELEGRAM RECEIVED.

FROM 2nd from London # 5747.

by *Mark A. Eckhoff*
Date *Oct 27, 1958*

"We intend to begin on the first of February unrestricted submarine warfare. We shall endeavor in spite of this to keep the United States of America neutral. In the event of this not succeeding, we make Mexico a proposal of alliance on the following basis: make war together, make peace together, generous financial support and an understanding on our part that Mexico is to reconquer the lost territory in Texas, New Mexico, and Arizona. The settlement in detail is left to you. You will inform the President of the above most secretly as soon as the outbreak of war with the United States of America is certain and add the suggestion that he should, on his own initiative, ~~write~~ ^{invite} Japan to immediate adherence and at the same time mediate between Japan and ourselves. Please call the President's attention to the fact that the ruthless employment of our submarines now offers the prospect of compelling England in a few months to make peace." Signed, ZIMMERMANN.

Zimmermann's gamble



Arthur Zimmermann, German foreign secretary in 1916.

- Zimmermann publicly admitted in March 1917 that it was genuine.
- The publishing swayed US opinion to declare war only weeks later.

The Enigma

- A polyalphabetic substitution cipher
 - Three scramblers in series.
 - An additional reflector.
 - A plugboard that allows the user to additionally swap up to 6 characters.
- Frequency analysis is effectively useless.
- Reversible!
- The breaking of the Enigma was not made public until 1974 – more than 30 years after the fact.



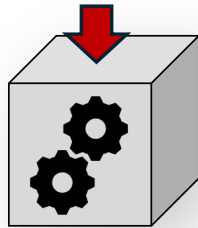


The advent of computers

Modern block ciphers

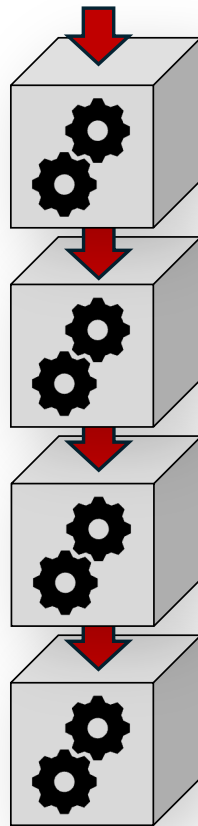
- With the adoption of computer communication, more complex block ciphers started emerging.
- In 1973 the DES algorithm, developed by IBM, was adopted as the national standard in the US.
- DES was, however, made deliberately weak with its 56-bit key.
- During a transitional period, triple-DES was considered adequate, although not great.
- AES has since become a globally accepted standard.

Anatomy of a modern block cipher



A substitution box (an “S-box”) processes the incoming data frame in a specific way.

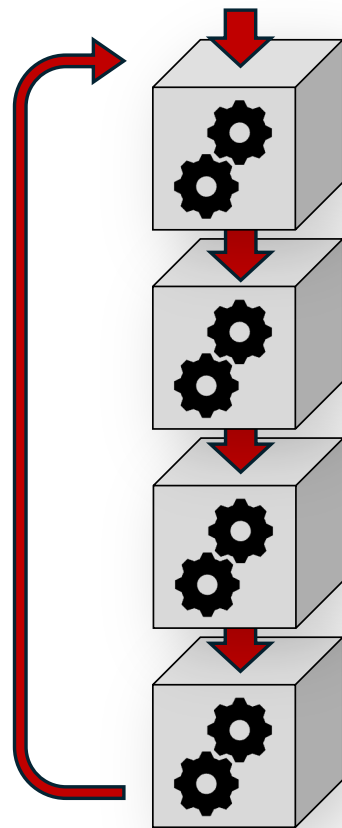
Anatomy of a modern block cipher



A substitution box (an “S-box”) processes the incoming data frame in a specific way.

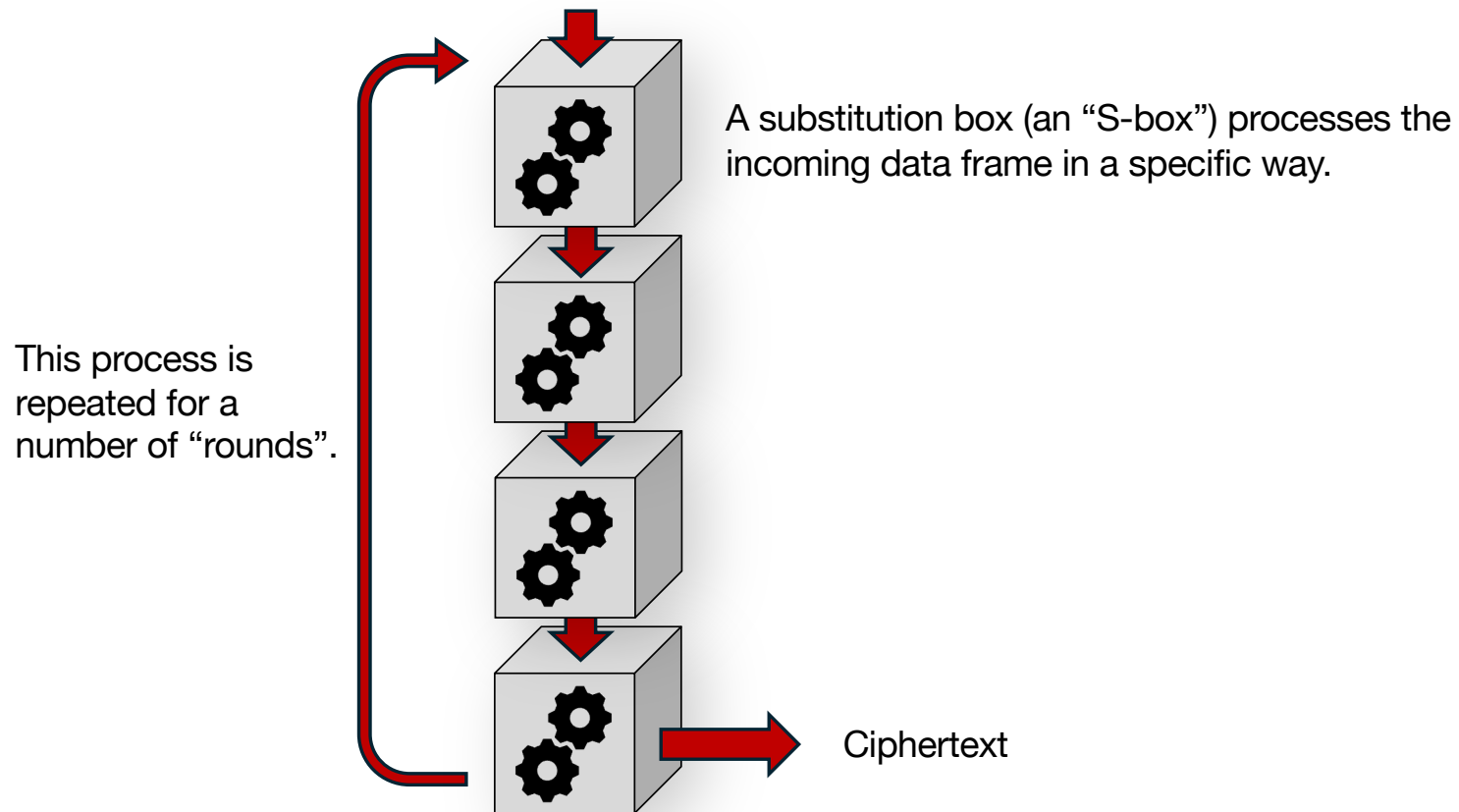
Anatomy of a modern block cipher

This process is repeated for a number of “rounds”.



A substitution box (an “S-box”) processes the incoming data frame in a specific way.

Anatomy of a modern block cipher



AES

Plaintext:

Hex:

A	t	t	a	c	k		a	t		d	a	w	n	!	
41	74	74	61	63	6B	20	61	74	20	64	61	77	6E	21	00

Password:

Hex:

i	k	n	o	w	w	h	a	t	y	o	u	d	i	d	!
69	6B	6E	6F	77	77	68	61	74	79	6F	75	64	69	64	21

AES

Data frame:

41	74	74	61
63	6b	20	61
74	20	64	61
77	6e	21	00



Plaintext:

A	t	t	a	c	k		a	t		d	a	w	n	!	
41	74	74	61	63	6B	20	61	74	20	64	61	77	6E	21	00

Hex:

Password:

i	k	n	o	w	w	h	a	t	y	o	u	d	i	d	!
69	6B	6E	6F	77	77	68	61	74	79	6F	75	64	69	64	21

Hex:

AES

Data frame:

41	74	74	61
63	6b	20	61
74	20	64	61
77	6e	21	00

Key 0:

69	77	74	64
6b	77	79	69
6e	68	6f	64
6f	61	75	21

Key 1:

91	E6	92	F6
28	5f	26	4f
93	Fb	94	F0
2c	4d	38	19

Key 2:

17	f1	63	95
a4	fb	dd	92
47	bc	28	d8
6e	23	1b	02

Key 3:

5c	ac
c5	3e
30	8c
44	67

Key expansion

Turns a key into 10, 12 or 14 “round keys”, 16 bytes each.

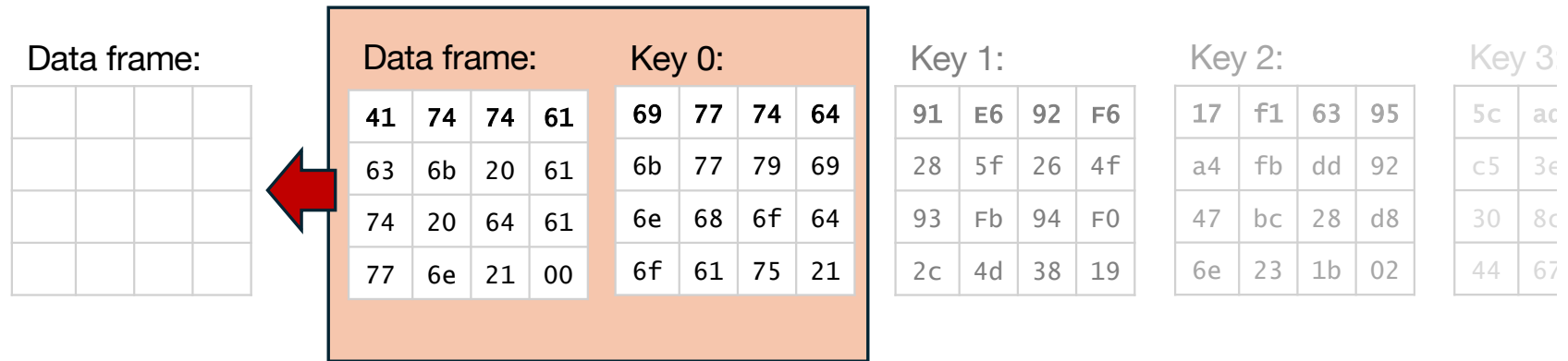
Password:

Hex:

i	k	n	o	w	w	h	a	t	y	o	u	d	i	d	!
69	6B	6E	6F	77	77	68	61	74	79	6F	75	64	69	64	21



AES



Key expansion

Turns a key into 10, 12 or 14 “round keys”, 16 bytes each.

AddRoundKey

XORs the data frame with the round key

SubBytes

ShiftRows

MixColumns

AES

Data frame:

Data frame:

41	74	74	61
63	6b	20	61
74	20	64	61
	21	00	

Key 0:

69	77	74	64
6b	77	79	69
6e	68	6f	64
6f	61	75	21

Key 1:

91	E6	92	F6
28	5f	26	4f
93	Fb	94	F0
2c	4d	38	19

Key 2:

17	f1	63	95
a4	fb	dd	92
47	bc	28	d8
6e	23	1b	02

Key 3:

5c	ad
c5	3e
30	8c
44	67

XOR: Either one or the other, but not both.

0x41

0x69

= 0x28

Key expansion turns a 128-bit key into 12 or 14 "round keys", 16 bytes each.

AddRoundKey

XORs the data frame with the round key

SubBytes

ShiftRows

MixColumns

AES

Data frame:

Data frame:

41	74	74	61
63	6b	20	61
74	20	64	61
21	00		

Key 0:

69	77	74	64
6b	77	79	69
6e	68	6f	64
6f	61	75	21

Key 1:

91	E6	92	F6
28	5f	26	4f
93	Fb	94	F0
2c	4d	38	19

Key 2:

17	f1	63	95
a4	fb	dd	92
47	bc	28	d8
6e	23	1b	02

Key 3:

5c	ad
c5	3e
30	8c
44	67

XOR: Either one or the other, but not both.

0x41 0 1 0 0 0 0 0 1

0x69 0 1 1 0 1 0 0 1

= 0x28

Key expansion turns a 128-bit key into 12 or 14 "round keys", 16 bytes each.

AddRoundKey

XORs the data frame with the round key

SubBytes

ShiftRows

MixColumns

AES

Data frame:

Data frame:

41	74	74	61
63	6b	20	61
74	20	64	61
21	00		

Key 0:

69	77	74	64
6b	77	79	69
6e	68	6f	64
6f	61	75	21

Key 1:

91	E6	92	F6
28	5f	26	4f
93	Fb	94	F0
2c	4d	38	19

Key 2:

17	f1	63	95
a4	fb	dd	92
47	bc	28	d8
6e	23	1b	02

Key 3:

5c	ad
c5	3e
30	8c
44	67

XOR: Either one or the other, but not both.

0x41 0 1 0 0 0 0 0 1

0x69 0 1 1 0 1 0 0 1

= **0x28** 0 0 1 0 1 0 0 0

Key expansion turns a 128-bit key into 12 or 14 "round keys", 16 bytes each.

AddRoundKey

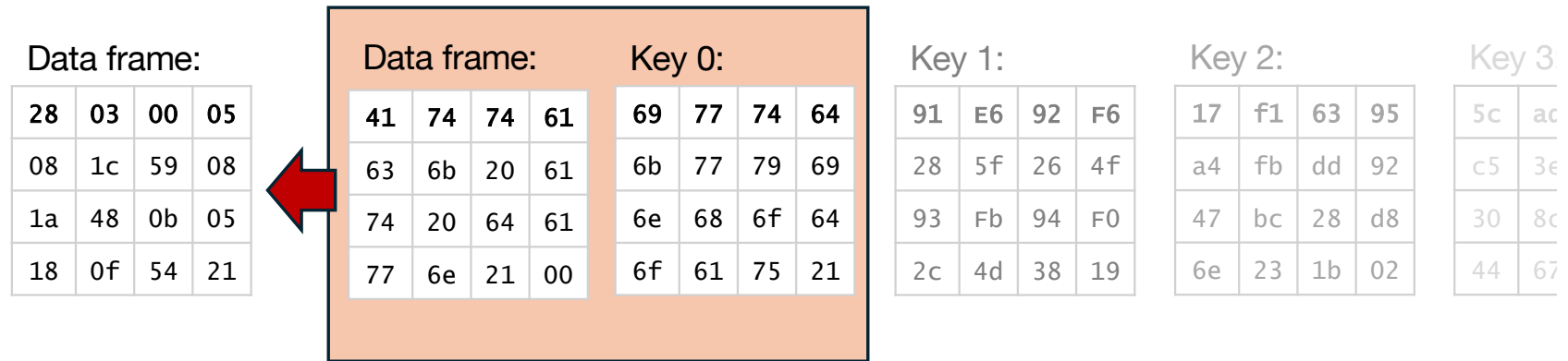
XORs the data frame with the round key

SubBytes

ShiftRows

MixColumns

AES



Key expansion

Turns a key into 10, 12 or 14 “round keys”, 16 bytes each.

AddRoundKey

XORs the data frame with the round key

SubBytes

ShiftRows

MixColumns

AES

Data frame:

34	7b	63	6b
30	9c	cb	30
a2	52	2b	6b
ad	76	20	fd



63	7c	77	7b	f2	6b	6f	c5	30	01	67	2b	fe	d7	ab	76
ca	82	c9	7d	fa	59	47	f0	ad	d4	a2	af	9c	a4	72	c0
b7	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	15
04	c7	23	c3	18	96	05	9a	07	12	80	e2	eb	27	b2	75
09	83	2c	1a	1b	6e	5a	a0	52	3b	d6	b3	29	e3	2f	84
53	d1	00	ed	20	fc	b1	5b	6a	cb	be	39	4a	4c	58	cf
d0	ef	aa	fb	43	4d	33	85	45	f9	02	7f	50	3c	9f	a8
51	a3	40	8f	92	9d	38	f5	bc	b6	da	21	10	ff	f3	d2
cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	73
60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	db
e0	32	3a	0a	49	6	24	5c	c2	d3	ac	62	91	95	e4	79
e7	c8	37	6d	8d	d5	4e	a9	6c	56	f4	ea	65	7a	ae	08
ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8b	8a
70	3e	b5	66	48	03	f6	0e	61	35	57	b9	86	c1	1d	9e
e1	f8	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	df
8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb	16

Key 1:

91	E6	92	F6
28	5f	26	4f
93	Fb	94	F0
2c	4d	38	19

Key 2:

17	f1	63	95
a4	fb	dd	92
47	bc	28	d8
6e	23	1b	02

Key 3:

5c	ac
c5	3e
30	8c
44	67

Key expansion

Turns a key into 10, 12 or 14 “round keys”, 16 bytes each.

AddRoundKey

XORs the data frame with the round key

SubBytes

Applies a substitution (S-box) using a static table

ShiftRows

MixColumns

AES

Data frame:

34	7b	63	6b
9c	cb	30	30
2b	6b	a2	52
fd	ad	76	20



Data frame:

			34	7b	63	6b	
			30	9c	cb	30	
		a2	52	2b	6b		
ad	76	20	fd				

Key 1:

91	E6	92	F6
28	5f	26	4f
93	Fb	94	F0
2c	4d	38	19

Key 2:

17	f1	63	95
a4	fb	dd	92
47	bc	28	d8
6e	23	1b	02

Key 3:

5c	ac
c5	3e
30	8c
44	67

Key expansion

Turns a key into 10, 12 or 14 “round keys”, 16 bytes each.

AddRoundKey

XORs the data frame with the round key

SubBytes

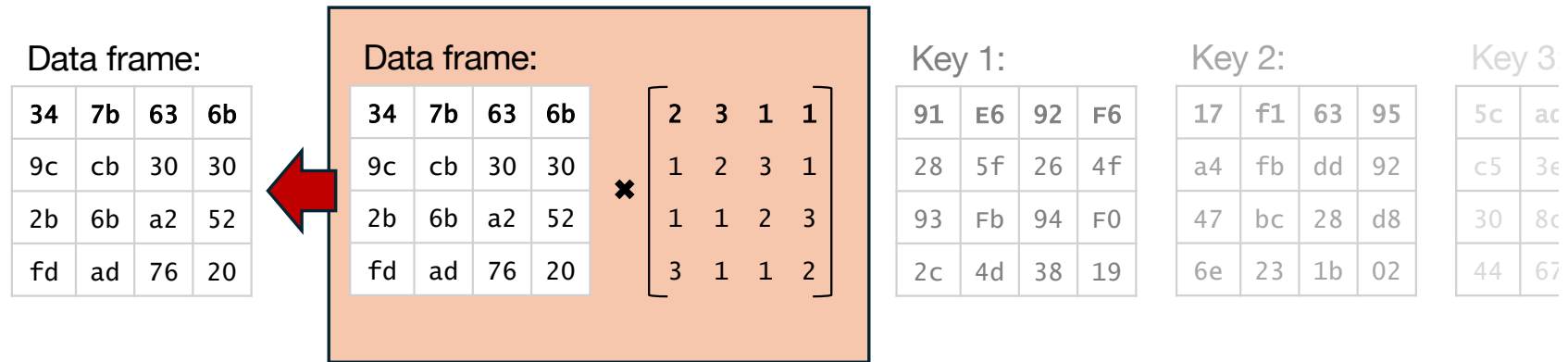
Applies a substitution (S-box) using a static table

ShiftRows

Shifts the rows in the data frame

MixColumns

AES



Key expansion

Turns a key into 10, 12 or 14 “round keys”, 16 bytes each.

AddRoundKey

XORs the data frame with the round key

SubBytes

Applies a substitution (S-box) using a static table

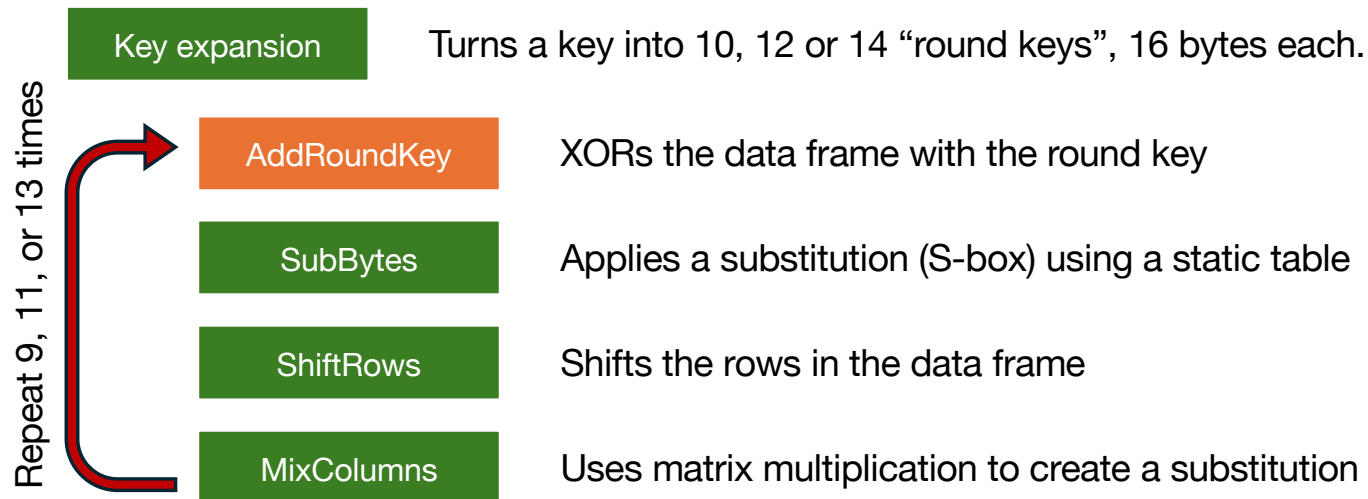
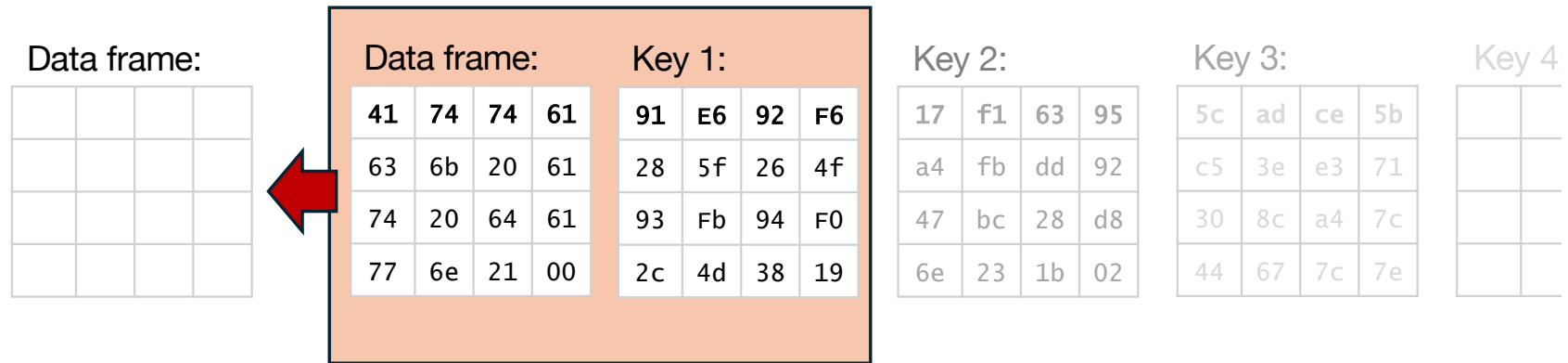
ShiftRows

Shifts the rows in the data frame

MixColumns

Uses matrix multiplication to create a substitution

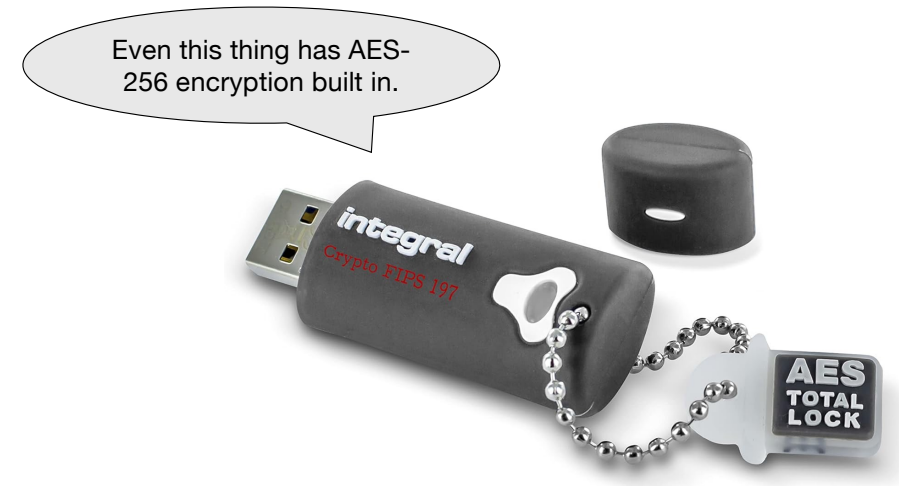
AES



AES

Used for:

- Data encryption for TLS (formerly SSL)
- Full disk encryption
- AES is ubiquitous - found in
 - software,
 - operating systems,
 - CPUs (including with native instructions),
 - PLCs, storage controllers, etc
 - USB flash drives
- The design allows for impressively high throughput



Hashing

Encryption

- Encryption generates “ciphertext”.

Please pay the recipient \$100.



09urxmo298r2eofwehfi2ehfoi2efh

Hashing

- Hashing creates a “digest”.

Please pay the recipient \$100.



0x4273a48237e9280c

Hashing

Encryption

- Encryption generates “ciphertext”.
- Encryption is *reversible*. One ciphertext generates exactly one plaintext.

Please pay the recipient \$100.



09urxmo298r2eofwehfi2ehfoi2efh

Hashing

- Hashing creates a “digest”.
- Hashing is *not reversible*. Sometimes, the two plaintexts can generate the same digest

Please pay the recipient \$100.



0x4273a48237e9280c

Hashing

Encryption

- Encryption generates “ciphertext”.
- Encryption is *reversible*. One ciphertext generates exactly one plaintext.
- Encryption uses a key or password

Please pay the recipient \$100.



09urxmo298r2eofwehfi2ehfoi2efh

Hashing

- Hashing creates a “digest”.
- Hashing is *not reversible*. Sometimes, the two plaintexts can generate the same digest
- Hashing can use a “salt”

Please pay the recipient \$100.



0x4273a48237e9280c

Hashing

Encryption

- Encryption generates “ciphertext”.
- Encryption is *reversible*. One ciphertext generates exactly one plaintext.
- Encryption uses a key or password
- Length of ciphertext is similar in length to plaintext

Please pay the recipient \$100.



09urxmo298r2eofwehfi2ehfoi2efh

Hashing

- Hashing creates a “digest”.
- Hashing is *not reversible*. Sometimes, the two plaintexts can generate the same digest
- Hashing can use a “salt”
- Digests are typically fixed-width

Please pay the recipient \$100.



0x4273a48237e9280c

Hashing

Hashing

passw0rd#1!



0x138d20ec87452a0f

Hashing

Hashing

passw0rd#1!



0x138d20ec87452a0f

0x138d20ec87452a0f



passw0rd#1!



There are “**rainbow tables**” for common passwords, allowing an attacker to reverse-engineer hashes.

Hashing

Hashing

passw0rd#1!



0x138d20ec87452a0f



0x138d20ec87452a0f



passw0rd#1!

Hashing with salt

Salt

xyzpassw0rd#1!



0x39ac862c958d3460

Hashing

Hashing

passw0rd#1!



0x138d20ec87452a0f

0x138d20ec87452a0f



passw0rd#1!



Hashing with salt

Salt

xyzpassw0rd#1!



0x39ac862c958d3460

Different salt

abcpassw0rd#1!

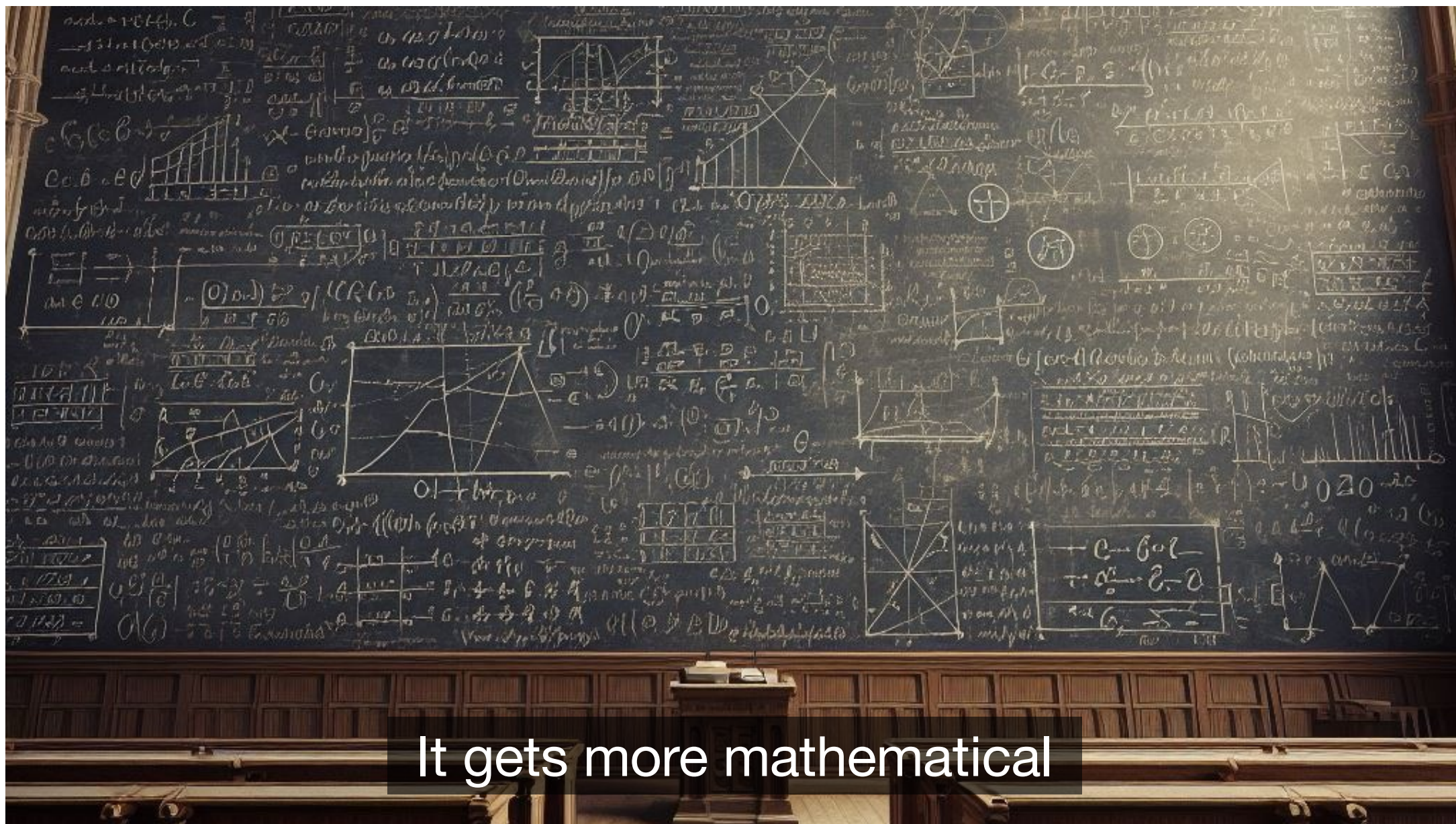


0xaa1209752f83942b

Hashing

How are hashes used?

- git commits
- Storing passwords
- Detecting changes in code/text, etc
- Digital signatures



It gets more mathematical

Asymmetric encryption

Also called “public key encryption”.

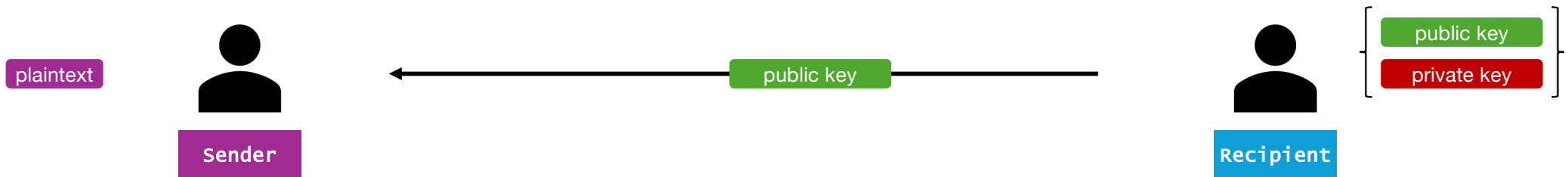
The two most common algorithms are

- RSA (Rivest, Shamir, Adleman)
- Elliptic Curve Encryption (ECC)

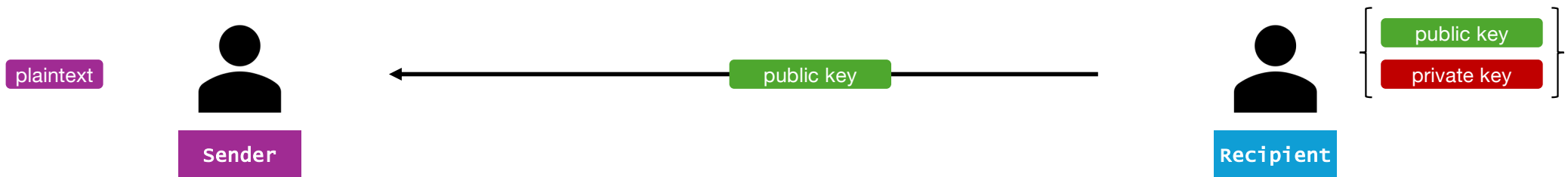
Asymmetric encryption



Asymmetric encryption

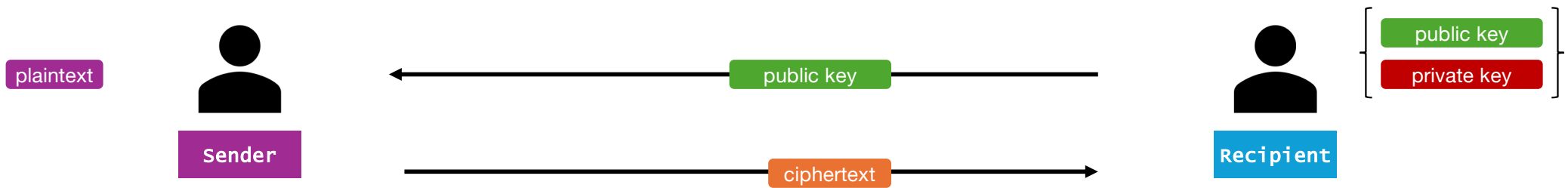


Asymmetric encryption



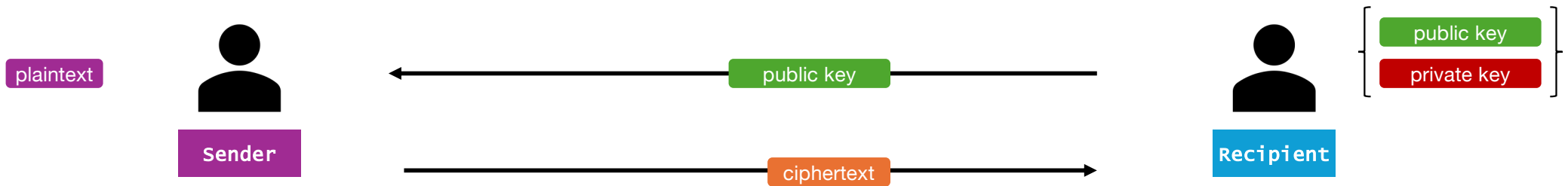
$$\text{ciphertext} = fn(\text{plaintext}, \text{public key})$$

Asymmetric encryption



$$\text{ciphertext} = fn(\text{plaintext}, \text{public key})$$

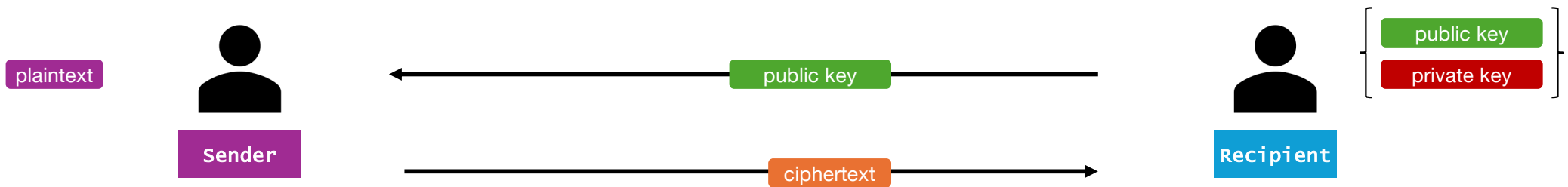
Asymmetric encryption



$$\text{ciphertext} = fn(\text{plaintext}, \text{public key})$$

$$\text{plaintext} = fn(\text{ciphertext}, \text{private key})$$

Asymmetric encryption



$$\text{ciphertext} = fn(\text{plaintext}, \text{public key})$$

$$\text{plaintext} = fn(\text{ciphertext}, \text{private key})$$

This "one-way" function is fairly simple math. But the reverse is very difficult.

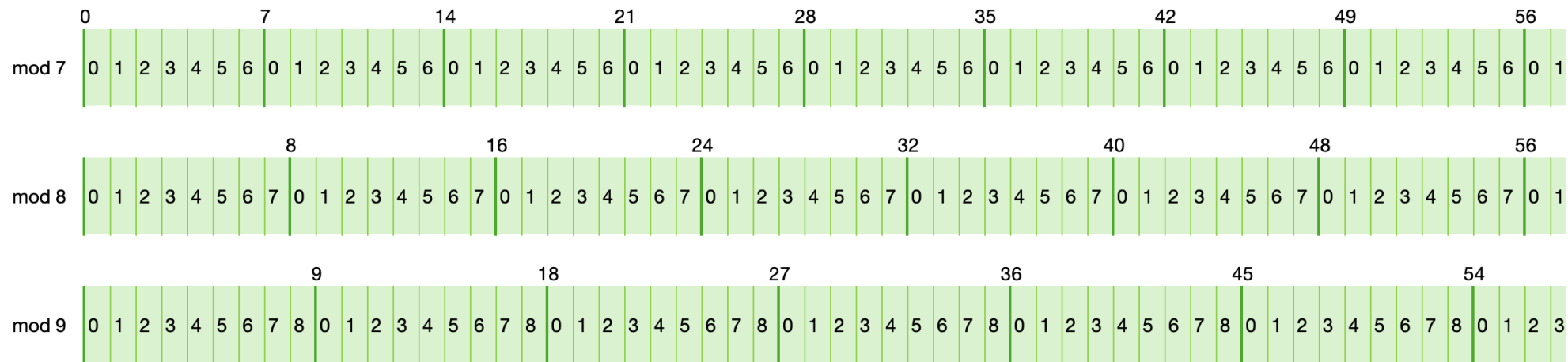
Jevons' number

"Can the reader say what two numbers multiplied together will produce the number 8,616,460,799? I think it unlikely that anyone but myself will ever know."

William Stanley Jevons
Principles of Science, 1874

Some required math basics

The **modulo** of an integer is the "remainder" when you divide the integer by the modulo.



Think of the modulo as an incrementing number that resets at intervals.

Some required math basics

A **prime** number is an integer that is evenly divisible by only itself and 1.

1	1 is prime
2	2 is prime
3	3 is prime
4	$4 = 2 \cdot 2$
5	5 is prime
6	$6 = 2 \cdot 3$
7	7 is prime
8	$8 = 2 \cdot 2 \cdot 2$

9	$9 = 3 \cdot 3$
10	$10 = 2 \cdot 5$
11	11 is prime
12	$12 = 2 \cdot 2 \cdot 3$
13	13 is prime
14	$14 = 7 \cdot 2$
15	$15 = 5 \cdot 3$
16	$16 = 2 \cdot 2 \cdot 2 \cdot 2$

17	17 is prime
18	$18 = 2 \cdot 3 \cdot 3$
19	19 is prime
20	$20 = 2 \cdot 2 \cdot 5$
21	$21 = 3 \cdot 7$
22	$22 = 11 \cdot 2$
23	23 is prime
24	$24 = 2 \cdot 2 \cdot 2 \cdot 3$

Every non-prime integer is a multiple of two or more prime numbers.

RSA



RSA

$p = 247\,193$
 $q = 395\,767$

PRIVATE

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

$\lambda(p \cdot q) = \text{lcm}(p-1, q-1) = 48\,915\,094\,536$

The least common multiple of the two prime factors.

mod (p-1)



mod (q-1)



RSA

$p = 247\,193$
 $q = 395\,767$

PRIVATE

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

$\lambda(p \cdot q) = \text{lcm}(p-1, q-1) = 48\,915\,094\,536$

e is chosen -
must be coprime
with $\lambda(p \cdot q)$

$e = 65\,537$

PUBLIC

Two numbers are said to be coprime when they don't share any divisors greater than 1.

3 and 8 are coprime.

4 and 6 are not coprime, because both can be evenly divided by 2.

RSA

$p = 247\,193$
 $q = 395\,767$

PRIVATE

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

$\lambda(p \cdot q) = \text{lcm}(p-1, q-1) = 48\,915\,094\,536$

$e = 65\,537$

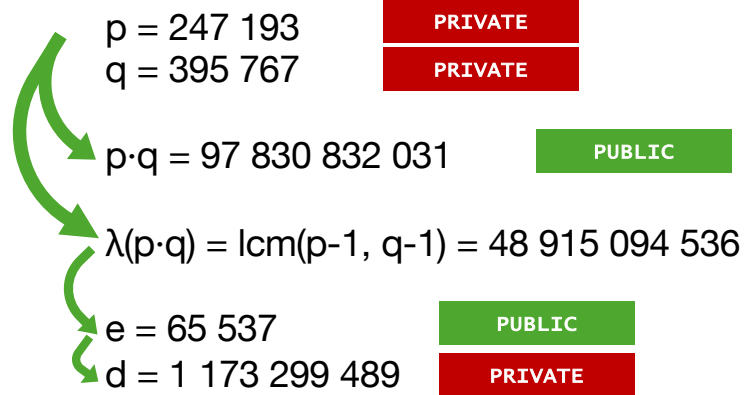
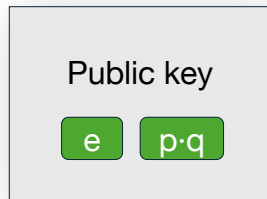
PUBLIC

$d = 1\,173\,299\,489$

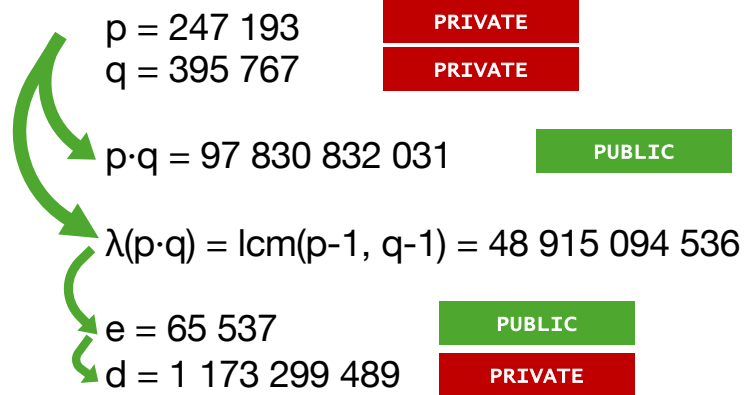
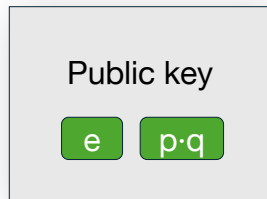
PRIVATE

d is a function of
 e and $\lambda(p \cdot q)$

RSA

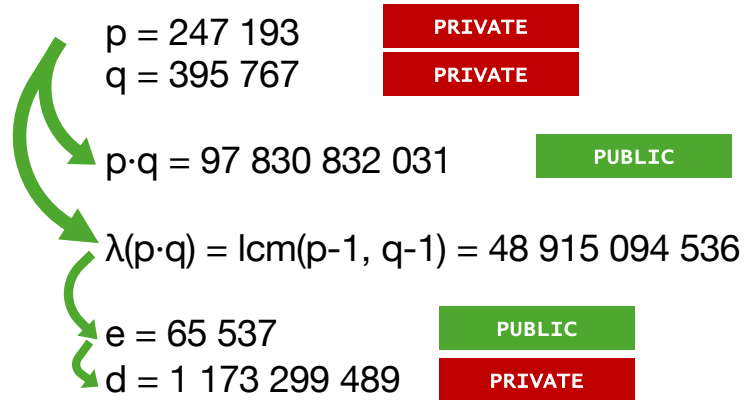
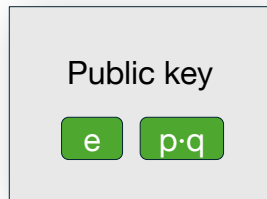


RSA



plaintext $m = 1\,234\,567$

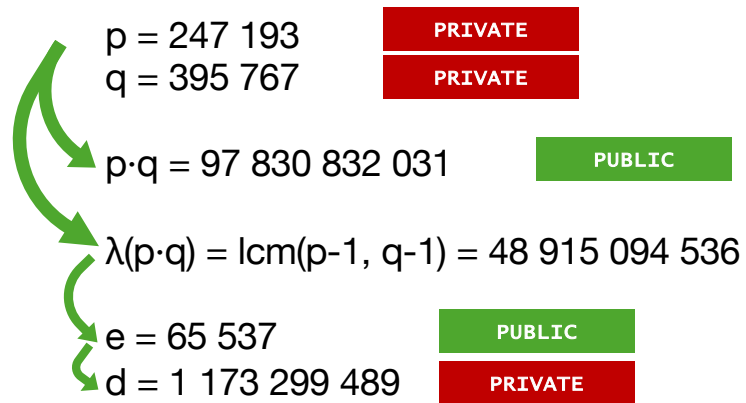
RSA



plaintext $m = 1\,234\,567$

Encryption: ciphertext $c = m^e \bmod p \cdot q = 65\,699\,382\,295$

RSA

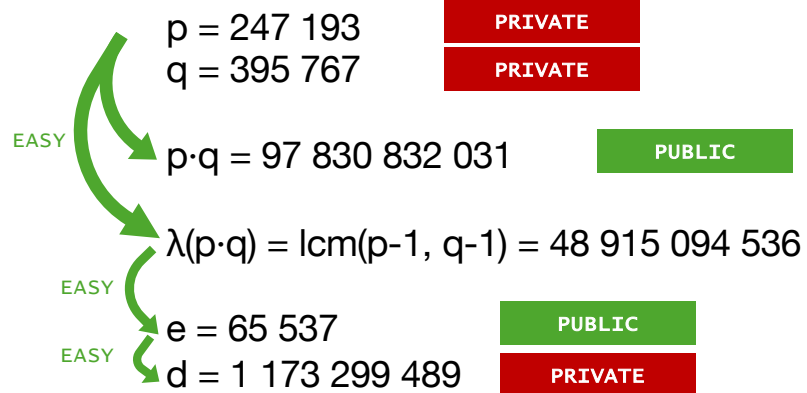


plaintext $m = 1\ 234\ 567$

Encryption: ciphertext $c = m^e \bmod p \cdot q = 65\ 699\ 382\ 295$

Decryption: plaintext $m = c^d \bmod p \cdot q = 1\ 234\ 567$

RSA

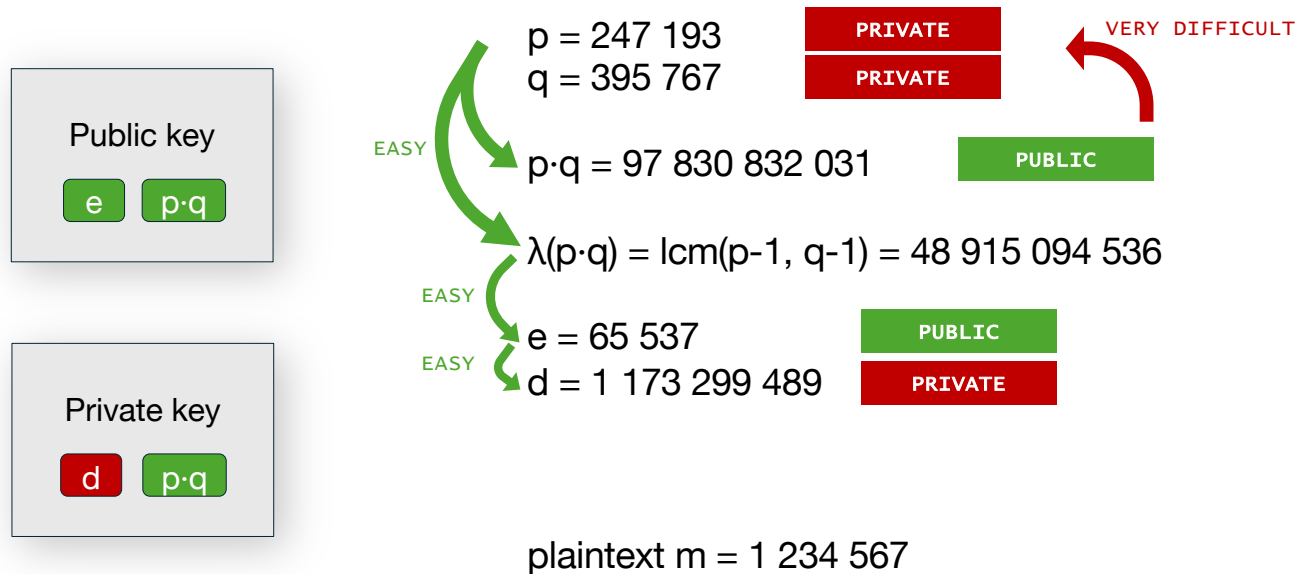


plaintext $m = 1\,234\,567$

Encryption: ciphertext $c = m^e \bmod p \cdot q = 65\,699\,382\,295$

Decryption: plaintext $m = c^d \bmod p \cdot q = 1\,234\,567$

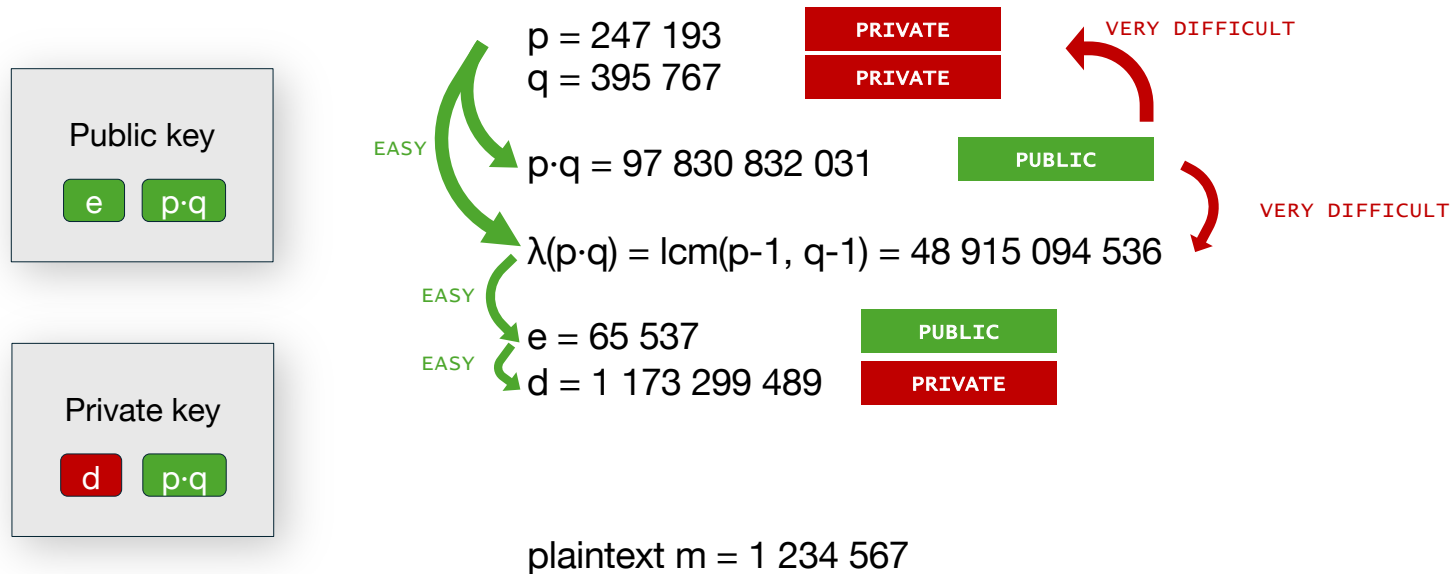
RSA



Encryption: ciphertext $c = m^e \bmod p \cdot q = 65\,699\,382\,295$

Decryption: plaintext $m = c^d \bmod p \cdot q = 1\,234\,567$

RSA



Encryption: ciphertext $c = m^e \bmod p \cdot q = 65\,699\,382\,295$

Decryption: plaintext $m = c^d \bmod p \cdot q = 1\,234\,567$

RSA

Which two prime numbers multiply to form this result?

$p \cdot q =$ 307 557 980 540 279 294 232 889 703 319 188 128 660 178 473 934 208 428 941 529 507 290 014 700 153 668 237 193 840 016 959 648 364
895 607 412 313 512 951 227 128 221 045 224 559 205 896 270 620 516 337 082 808 739 837 588 961 254 353 255 830 027 840 166 730 656 182
059 939 220 104 108 619 578 923 418 753 353 272 689 302 172 594 198 324 164 702 665 735 456 123 372 112 931 626 325 928 229 001 407 813
673 728 165 736 146 453 855 297 160 675 344 337 682 232 580 861 895 550 087 017 268 994 973 859 666 439 292 146 128 040 198 542 069 704
950 256 297 882 300 231 190 908 477 379 997 770 136 082 327 312 952 935 689 262 892 629 588 356 961 572 827 315 993 037 083 976 288 531
844 687 385 055 499 903 044 511 450 039 012 753 955 234 146 300 919 159 567 440 161 336 043 261 154 130 838 051 072 922 104 944 728 496
403 982 179 433 674 737 196 227 320 743 852 387 755 112 827 166 448 480 967 725 082 114 732 889 029 922 825 155 341 072 387 200 684 641
384 860 543 585 552 397 175 025 316 086 544 491 738 636 866 945 634 642 543 579 069 450 381 289 326 765 766 927 135 738 548 449 509 733
934 196 182 156 369 067 294 637 972 434 294 530 583 620 547 004 745 403 650 343 545 789 619 780 447 563 019 123 590 591

(roughly a 4096 bit key size)

RSA

Which two prime numbers

$p \cdot q = 307\ 557\ 980\ 540\ 279\ 294\ 895\ 607\ 412\ 313\ 512\ 951\ 227\ 12\ 059\ 939\ 220\ 104\ 108\ 619\ 578\ 92\ 673\ 728\ 165\ 736\ 146\ 453\ 855\ 29\ 950\ 256\ 297\ 882\ 300\ 231\ 190\ 90\ 844\ 687\ 385\ 055\ 499\ 903\ 044\ 51\ 403\ 982\ 179\ 433\ 674\ 737\ 196\ 22\ 384\ 860\ 543\ 585\ 552\ 397\ 175\ 02\ 934\ 196\ 182\ 156\ 369\ 067\ 294\ 63$

(roughly a 4096 bit key size)



You

Which two prime numbers can be multiplied to form 307 557 980 540 279 294 232 889 703 319 188 128 660 178 473 934 208 428 941 529 507 290 014 700 153 668 237 193 840 016 959 648 364 895 607 412 313 512 951 227 128 221 045 224 559 205 896 270 620 516 337 082 808 739 837 588 961 254 353 255 830 027 840 166 730 656 182 059 939 220 104 108 619 578 923 418 753 353 272 689 302 172 594 198 324 164 702 665 735 456 123 372 112 931 626 325 928 229 001 407 813 673 728 165 736 146 453 855 297 160 675 344 337 682 232 580 861 895 550 087 017 268 994 973 859 666 439 292 146 128 040 198 542 069 704 950 256 297 882 300 231 190 908 477 379 997 770 136 082 327 312 952 935 689 262 892 629 588 356 961 572 827 315 993 037 083 976 288 531 844 687 385 055 499 903 044 511 450 039 012 753 955 234 146 300 919 159 567 440 161 336 043 261 154 130 838 051 072 922 104 944 728 496 403 982 179 433 674 737 196 227 320 743 852 387 755 112 827 166 448 480 967 725 082 114 732 889 029 922 825 155 341 072 387 200 684 641 384 860 543 585 552 397 175 025 316 086 544 491 738 636

0 016 959 648 364 166 730 656 182 229 001 407 813 198 542 069 704 083 976 288 531 104 944 728 496 387 200 684 641 548 449 509 733

RSA

Which two prime numbers

$p \cdot q = 307\ 557\ 980\ 540\ 279\ 294\ 895\ 607\ 412\ 313\ 512\ 951\ 227\ 12\ 059\ 939\ 220\ 104\ 108\ 619\ 578\ 92\ 673\ 728\ 165\ 736\ 146\ 453\ 855\ 29\ 950\ 256\ 297\ 882\ 300\ 231\ 190\ 90\ 844\ 687\ 385\ 055\ 499\ 903\ 044\ 51\ 403\ 982\ 179\ 433\ 674\ 737\ 196\ 22\ 384\ 860\ 543\ 585\ 552\ 397\ 175\ 02\ 934\ 196\ 182\ 156\ 369\ 067\ 294\ 63$

(roughly a 4096 bit key size)



Copilot

The two prime numbers that can be multiplied to form the given number are:

$$\begin{array}{r} 5915587277 \\ \times \\ 519700308044732377919392591523925225 \end{array}$$

Please note that these are very large prime numbers and the multiplication of these two primes results in the given number. Prime numbers are numbers that have only two distinct positive divisors: 1 and itself. The concept of prime numbers is fundamental in number theory. They have several applications in computer science, cryptography, and other fields.



1 of 30

0 016 959 648 364
166 730 656 182
229 001 407 813
198 542 069 704
083 976 288 531
104 944 728 496
387 200 684 641
548 449 509 733

RSA

Which two prime numbers

$p \cdot q = 307\ 557\ 980\ 540\ 279\ 294\ 1$
895 607 412 313 512 951 227 12
059 939 220 104 108 619 578 92
673 728 165 736 146 453 855 29
950 256 297 882 300 231 190 90
844 687 385 055 499 903 044 51
403 982 179 433 674 737 196 22
384 860 543 585 552 397 175 02
934 196 182 156 369 067 294 63

(roughly a 4096 bit key size)



0 016 959 648 364
166 730 656 182
229 001 407 813
198 542 069 704
083 976 288 531
104 944 728 496
387 200 684 641
548 449 509 733

RSA

Which two prime numbers multiply to form this result?

$p \cdot q =$ 307 557 980 540 279 294 232 889 703 319 188 128 660 178 473 934 208 428 941 529 507 290 014 700 153 668 237 193 840 016 959 648 364 895 607 412 313 512 951 227 128 221 045 224 559 205 896 270 620 516 337 082 808 739 837 588 961 254 353 255 830 027 840 166 730 656 182 059 939 220 104 108 619 578 923 418 753 353 272 689 302 172 594 198 324 164 702 665 735 456 123 372 112 931 626 325 928 229 001 407 813 673 728 165 736 146 453 855 297 160 675 344 337 682 232 580 861 895 550 087 017 268 994 973 859 666 439 292 146 128 040 198 542 069 704 950 256 297 882 300 231 190 908 477 379 997 770 136 082 327 312 952 935 689 262 892 629 588 356 961 572 827 315 993 037 083 976 288 531 844 687 385 055 499 903 044 511 450 039 012 753 955 234 146 300 919 159 567 440 161 336 043 261 154 130 838 051 072 922 104 944 728 496 403 982 179 433 674 737 196 227 320 743 852 387 755 112 827 166 448 480 967 725 082 114 732 889 029 922 825 155 341 072 387 200 684 641 384 860 543 585 552 397 175 025 316 086 544 491 738 636 866 945 634 642 543 579 069 450 381 289 326 765 766 927 135 738 548 449 509 733 934 196 182 156 369 067 294 637 972 434 294 530 583 620 547 004 745 403 650 343 545 789 619 780 447 563 019 123 590 591

The correct answer is

$p =$ 5 357 543 035 931 336 604 742 125 245 300 009 052 807 024 058 527 668 037 218 751 941 851 755 255 624 680 612 465 991 894 078 479 290 637 973 364 587 765 734 125 935 726 428 461 570 217 992 288 787 349 287 401 967 283 887 412 115 492 710 537 302 531 185 570 938 977 091 076 523 237 491 790 970 633 699 383 779 582 771 973 038 531 457 285 598 238 843 271 083 830 214 915 826 312 193 418 602 834 035 927

and

$q =$ 57 406 534 763 712 726 211 641 660 058 884 099 201 115 885 104 434 760 023 882 136 841 288 313 069 618 515 692 832 974 315 825 313 495 922 298 231 949 373 138 672 355 948 043 152 766 571 296 567 808 332 659 269 564 994 572 656 140 000 344 389 574 120 022 435 714 463 495 031 743 122 390 807 731 823 194 181 973 658 513 020 233 176 985 452 498 279 081 199 404 472 314 802 811 655 824 768 082 110 985 166 340 672 084 454 492 229 252 801 189 742 403 957 029 450 467 388 250 214 501 358 353 312 915 261 004 066 118 140 645 880 633 941 658 603 299 497 698 209 063 510 889 929 202 021 079 926 591 625 770 444 716 951 045 960 277 478 891 794 836 019 580 040 978 608 315 291 377 690 212 791 863 007 764 174 393 209 716 027 254 457 637 891 941 312 587 717 764 400 411 421 385 408 982 726 881 092 425 574 515 033

RSA

Which two prime numbers multiply to form this result?

$p \cdot q =$ 307 557 980 540 279 29
895 607 412 313 512 951 227
059 939 220 104 108 619 578
673 728 165 736 146 453 855
950 256 297 882 300 231 190
844 687 385 055 499 903 044
403 982 179 433 674 737 196
384 860 543 585 552 397 175
934 196 182 156 369 067 294

The correct answer is

$p =$ 5 357 543 035 931 336 60
637 973 364 587 765 734 125
076 523 237 491 790 970 633

and

$q =$ 57 406 534 763 712 726 2
922 298 231 949 373 138 672 535 948 043 152 768 571 298 587 808 332 639 269 584 994 572 636 140 000 544 589 574 120 022 435 714 463 495
031 743 122 390 807 731 823 194 181 973 658 513 020 233 176 985 452 498 279 081 199 404 472 314 802 811 655 824 768 082 110 985 166 340
672 084 454 492 229 252 801 189 742 403 957 029 450 467 388 250 214 501 358 353 312 915 261 004 066 118 140 645 880 633 941 658 603 299
497 698 209 063 510 889 929 202 021 079 926 591 625 770 444 716 951 045 960 277 478 891 794 836 019 580 040 978 608 315 291 377 690 212
791 863 007 764 174 393 209 716 027 254 457 637 891 941 312 587 717 764 400 411 421 385 408 982 726 881 092 425 574 515 033

57 193 840 016 959 648 364
027 840 166 730 656 182
325 928 229 001 407 813
128 040 198 542 069 704
993 037 083 976 288 531
072 922 104 944 728 496
341 072 387 200 684 641
135 738 548 449 509 733
590 591

2 465 991 894 078 479 290
531 185 570 938 977 091
418 602 834 035 927

92 832 974 315 825 313 495



If you guessed correctly,
just shake your head and smile.
And never, ever tell a soul.

RSA

RSA has a published list of prime factors, some of which have prize money attached.

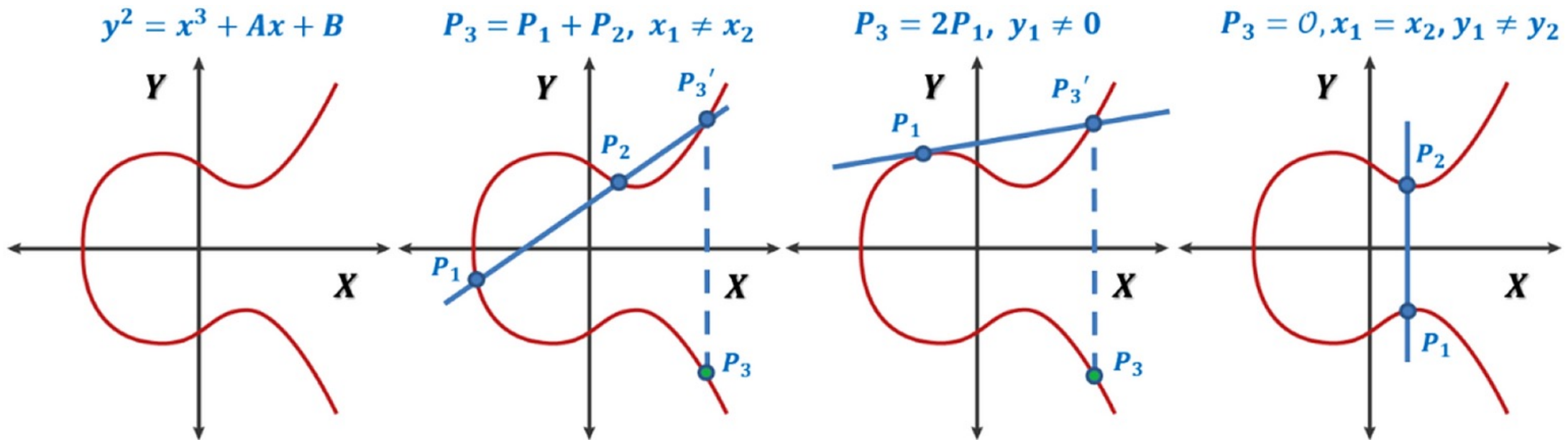
The largest RSA number factored so far is "RSA-250", equivalent of a 829 bit key, which was factored in 2020.

The computation effort is roughly equivalent to 2700 CPU core years on a 2.1 GHz Xeon.

The patent number for RSA, 4 405 829, is actually prime.

Elliptic Curve (ECC)

Way, *waaaaay* too complicated for an introductory session.



RSA & ECC: Asymmetric encryption

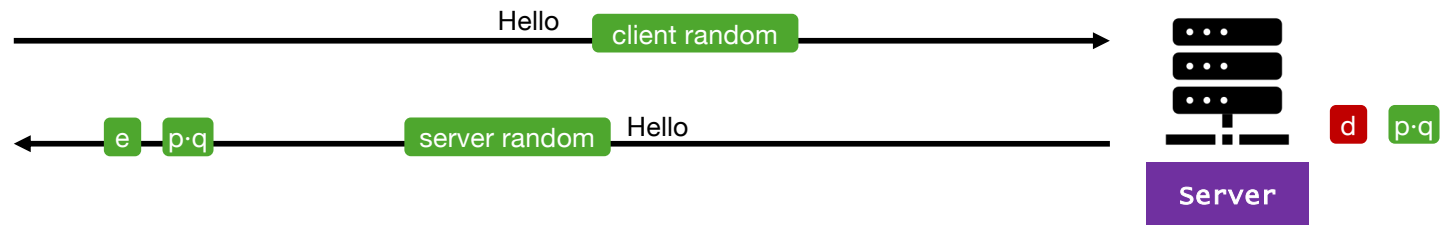
How is asymmetric encryption used?

- Key exchange for TLS (formerly SSL), SSH
- Blockchains, crypto currencies
- Secure signatures

Asymmetric algorithms are typically only used to encrypt the key or hash digest. The data is encrypted using much faster symmetric encryption.

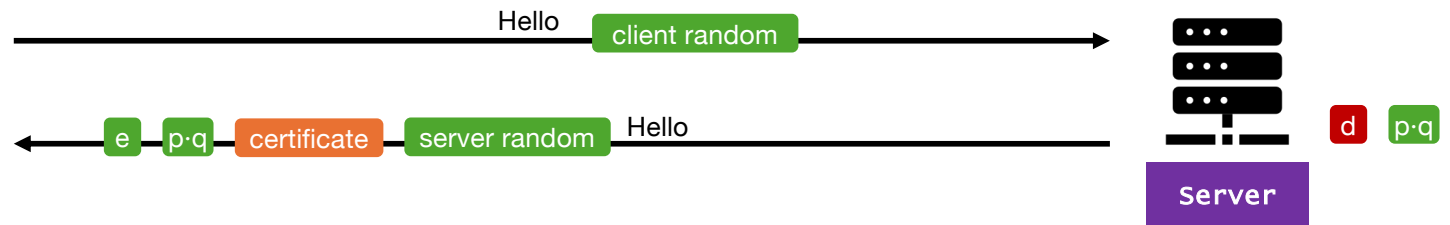
Big picture

Asymmetric



Big picture

Asymmetric



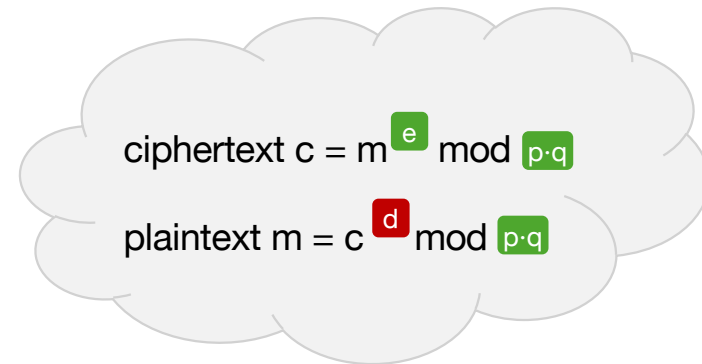
Big picture

Asymmetric



Signatures

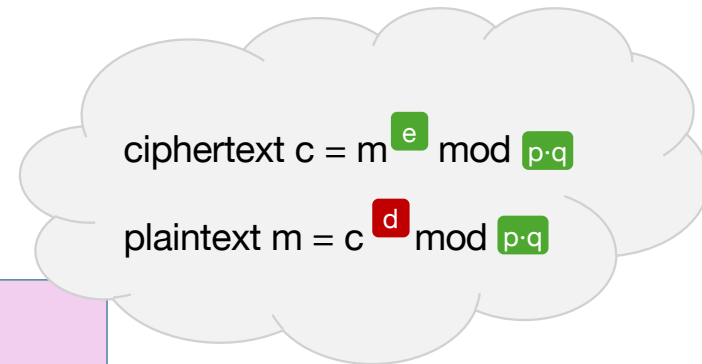
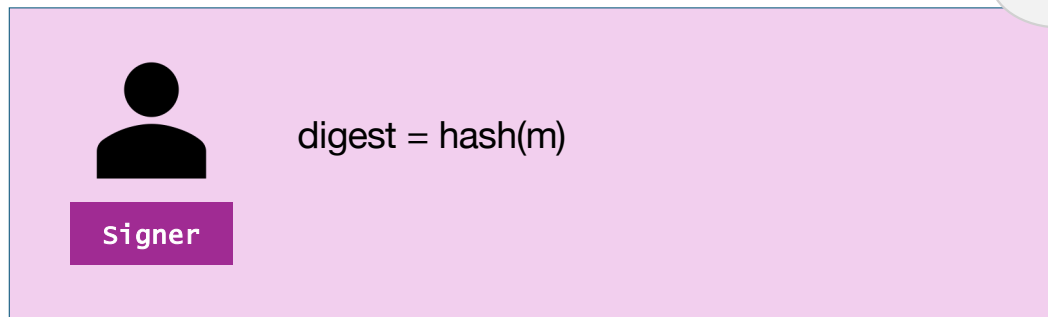
Signing uses the same function as encryption and decryption, but e and d are swapped.



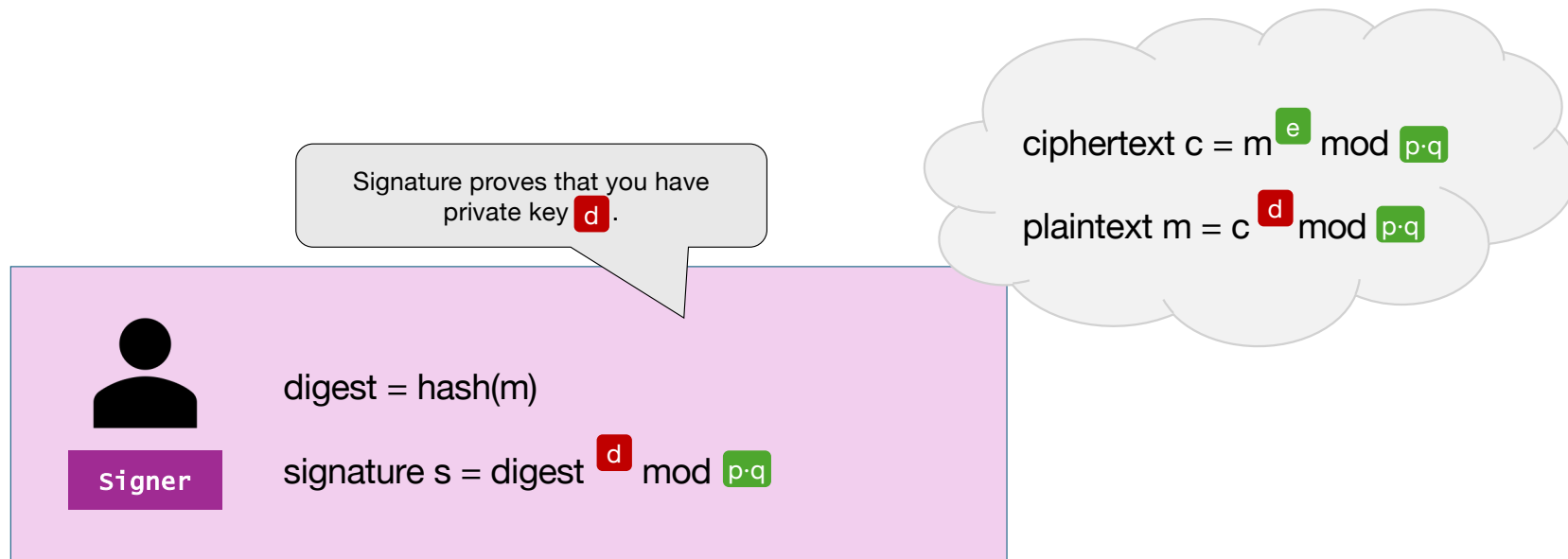
cloud

$$\text{ciphertext } c = m^e \bmod p \cdot q$$
$$\text{plaintext } m = c^d \bmod p \cdot q$$

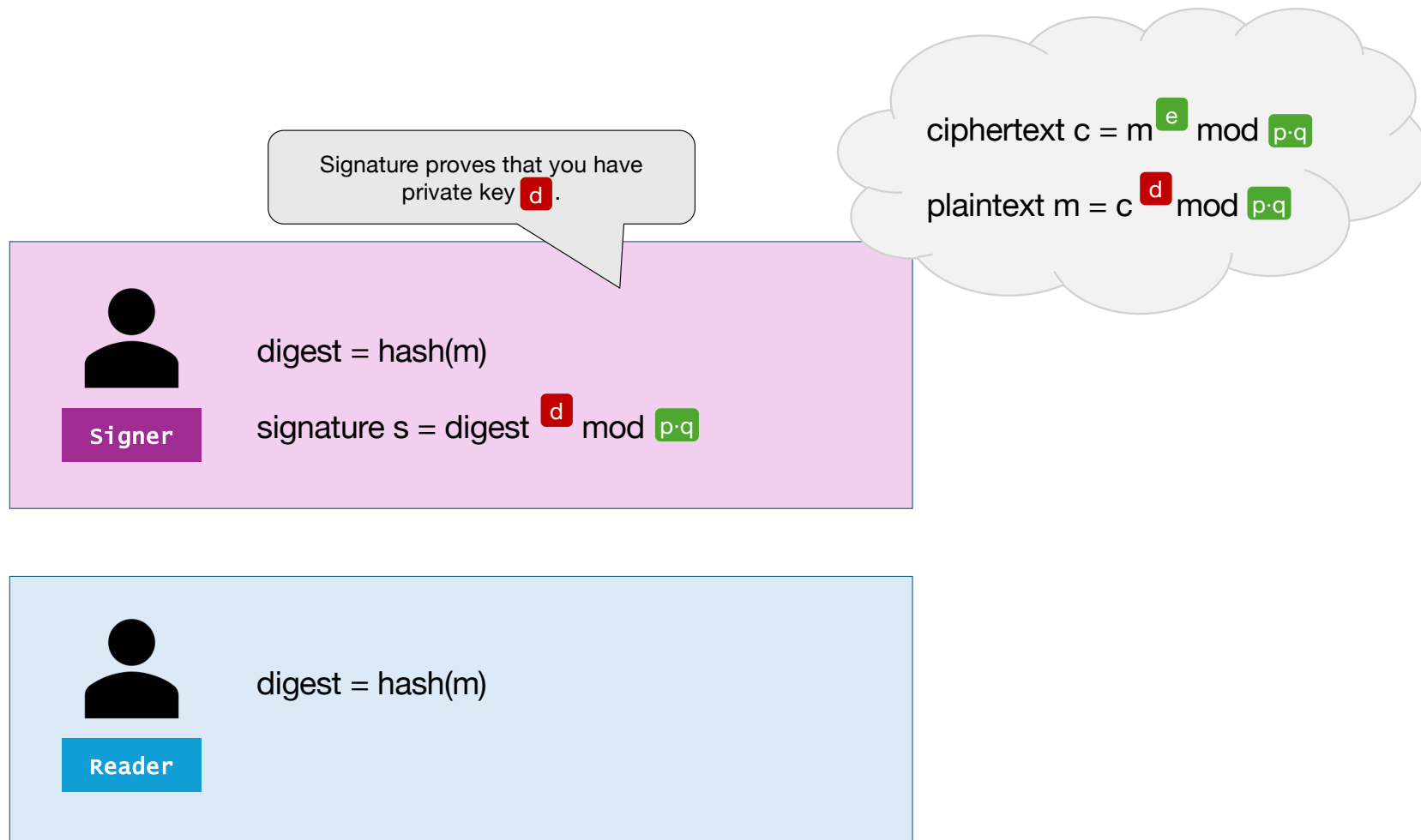
Signatures



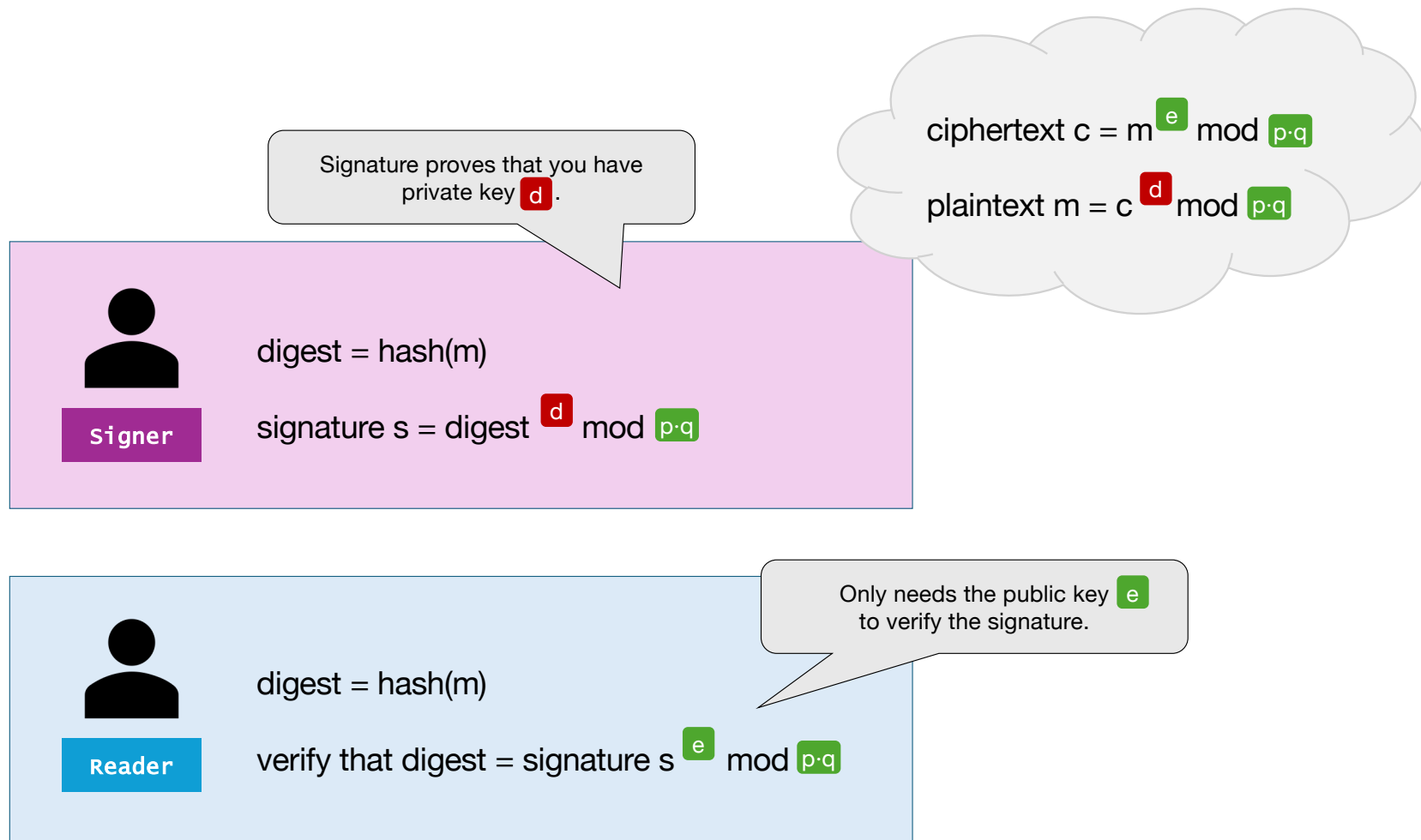
Signatures



Signatures



Signatures

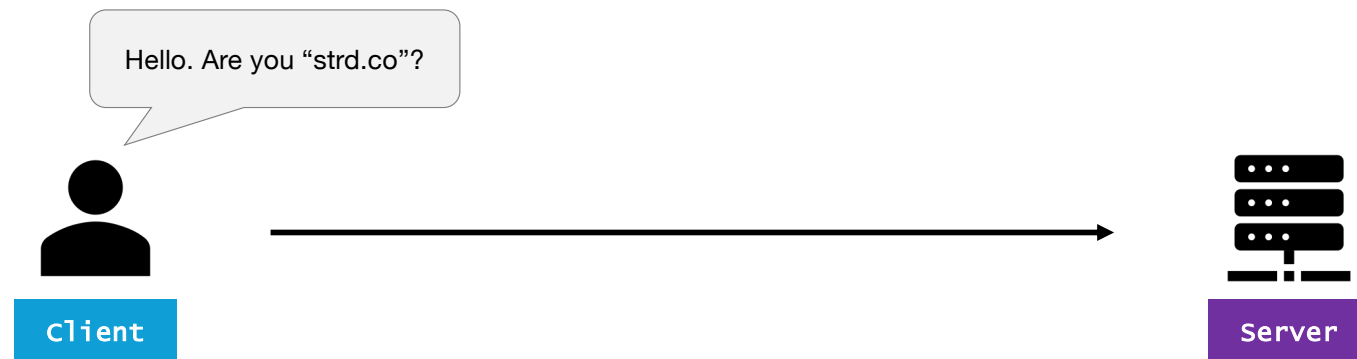


Signatures

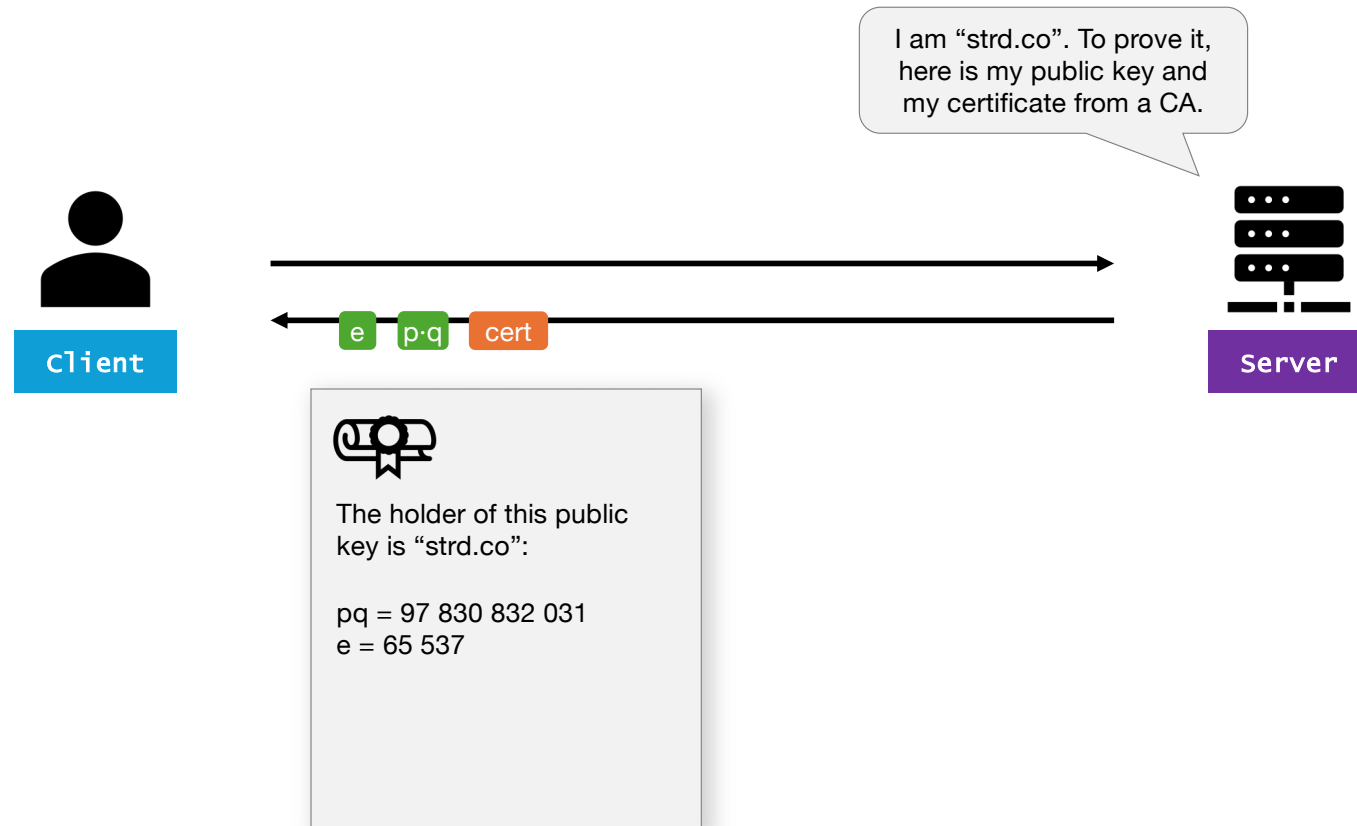
How are cryptographic signatures used?

- Code signing
- DNSSEC
- Signing certificates

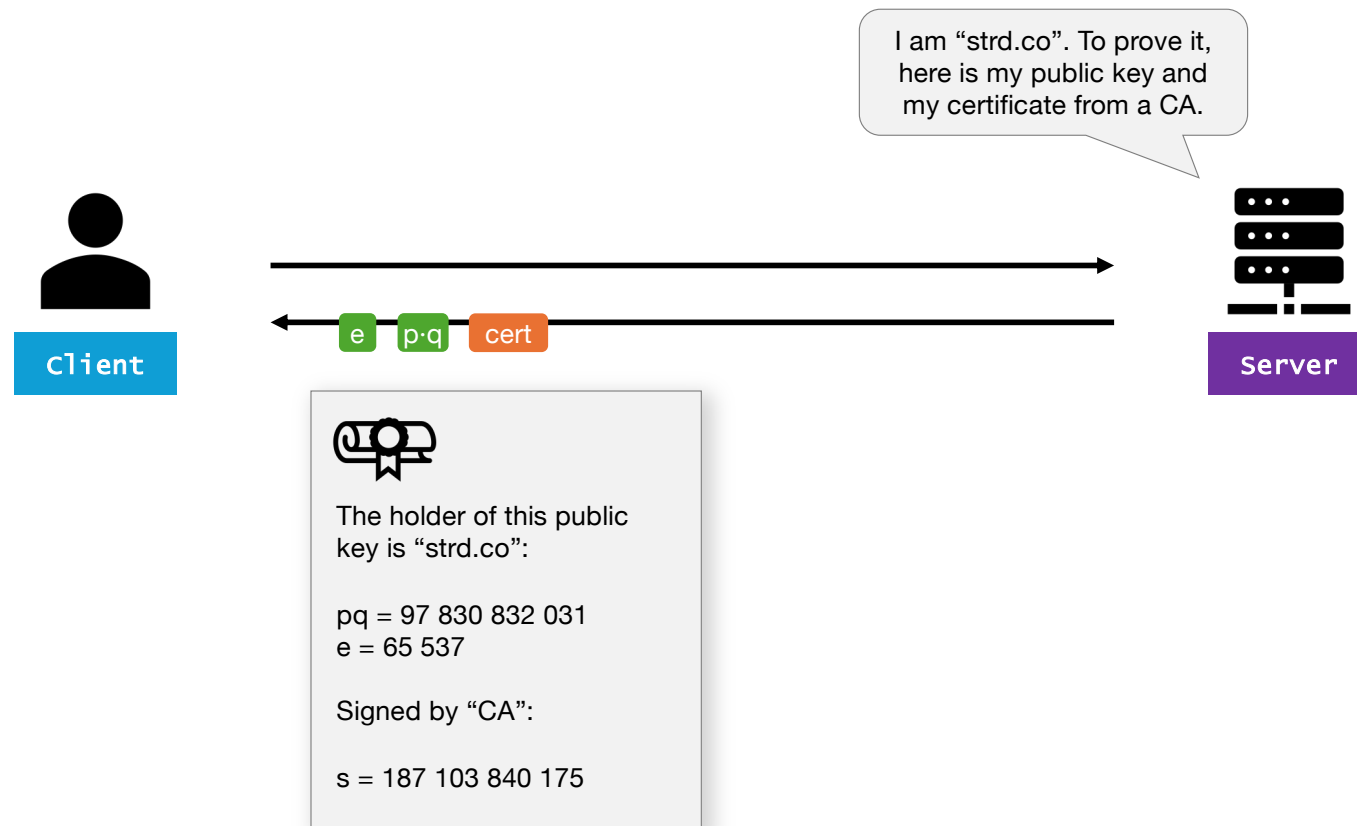
Certificate chains



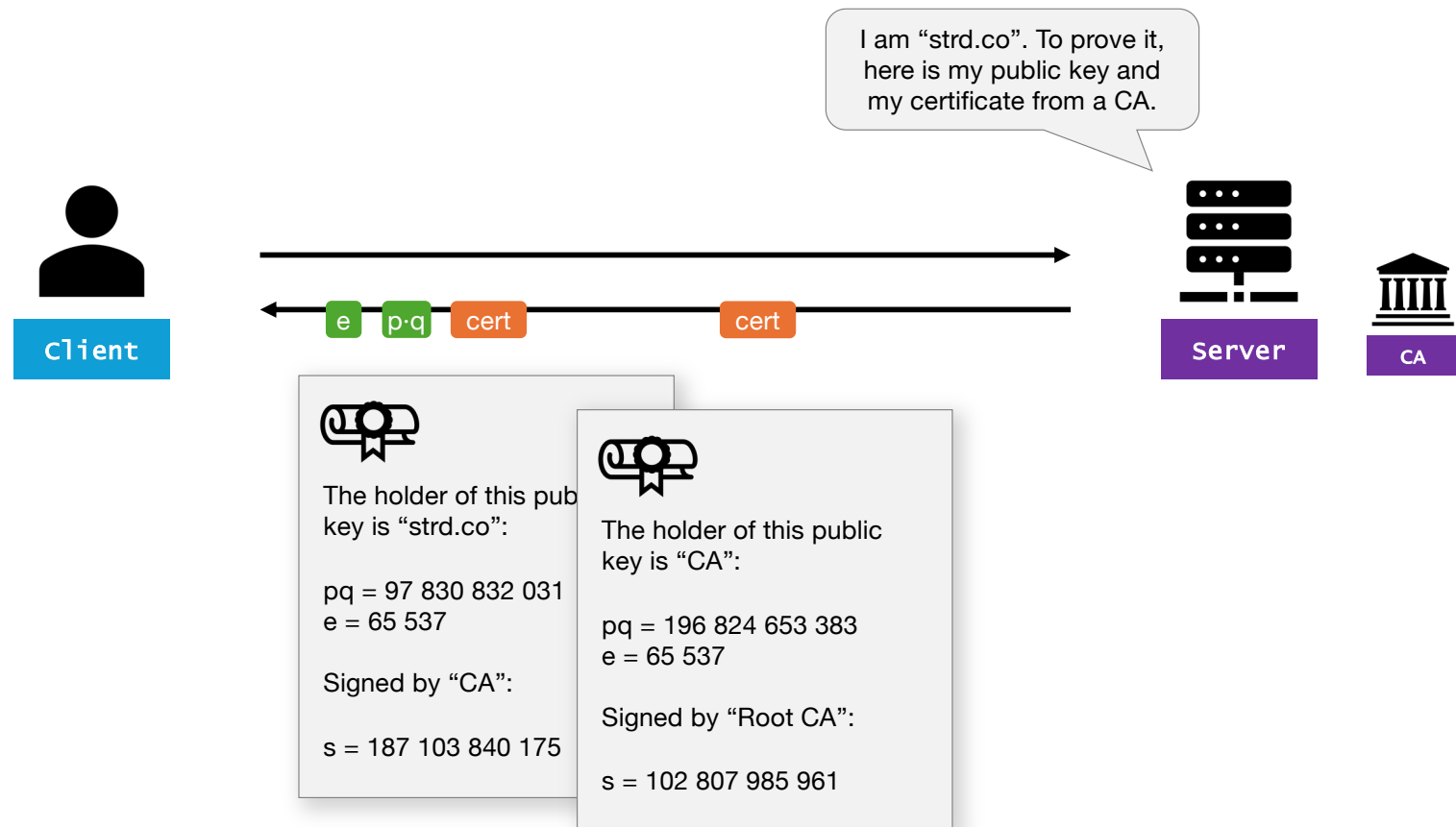
Certificate chains



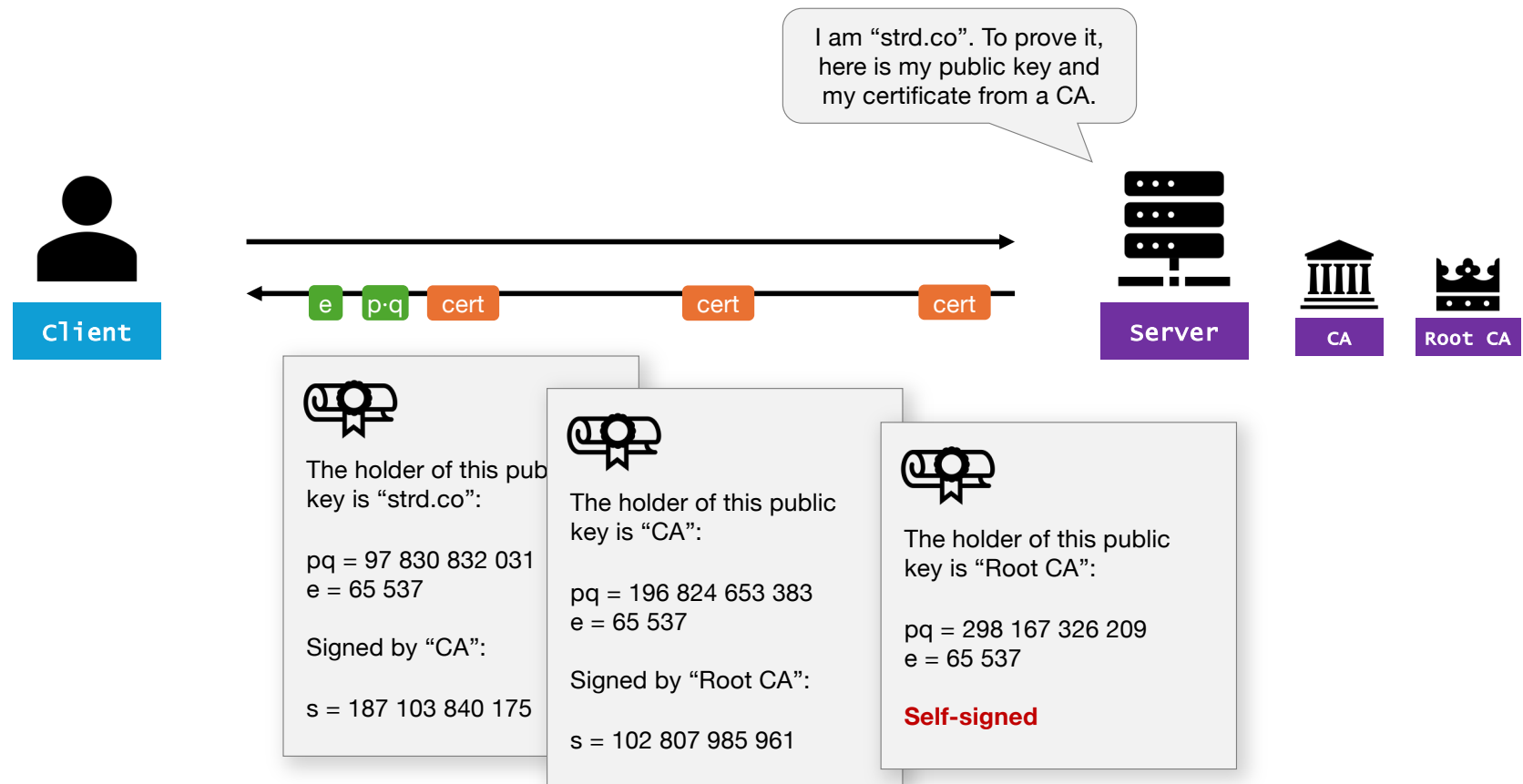
Certificate chains



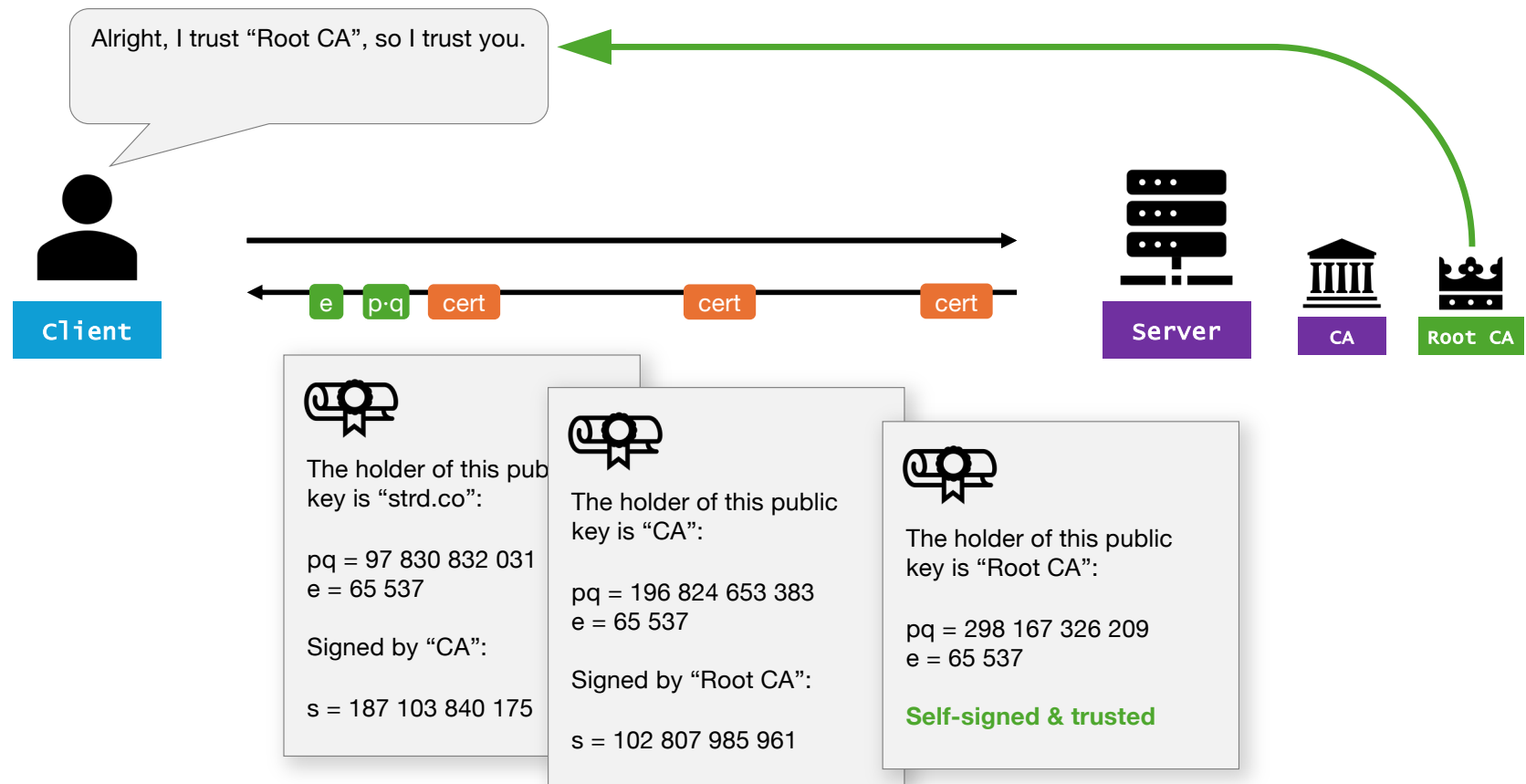
Certificate chains



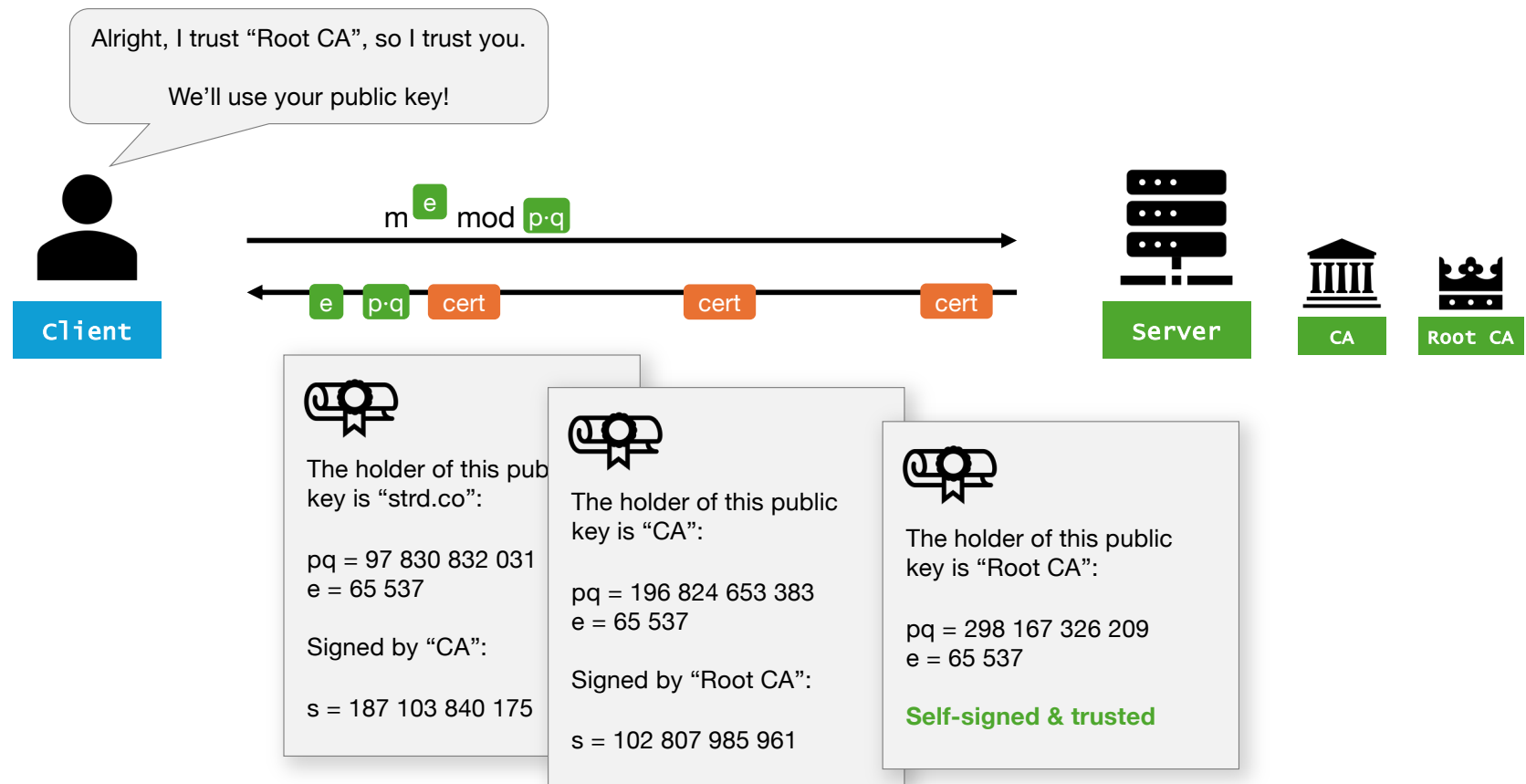
Certificate chains



Certificate chains



Certificate chains



Certificate chains

- A certificate is a “digital identity card”. The holder can prove that it has a matching private key using without publishing it.
- Just like identity cards, certificates have authorized issuers.
- Certificates can be self-issued (“self-signed”).
- Certificates mitigates man-in-the-middle (MITM) attacks.

Big picture

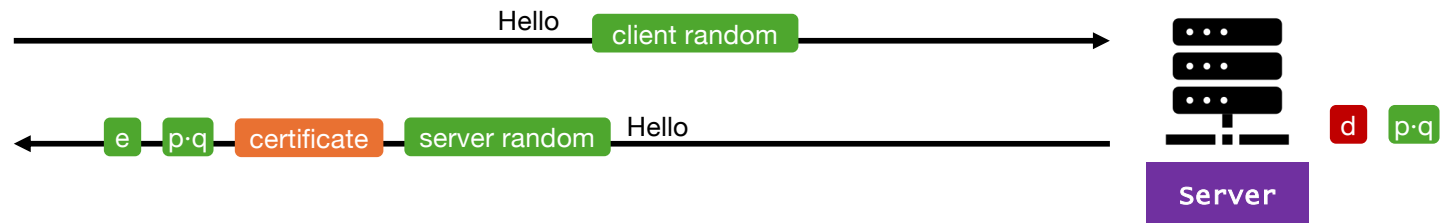
Asymmetric



Client

Validate

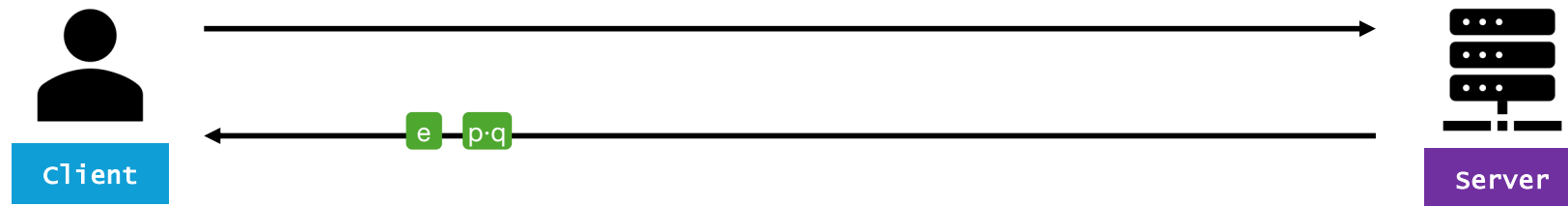
certificate



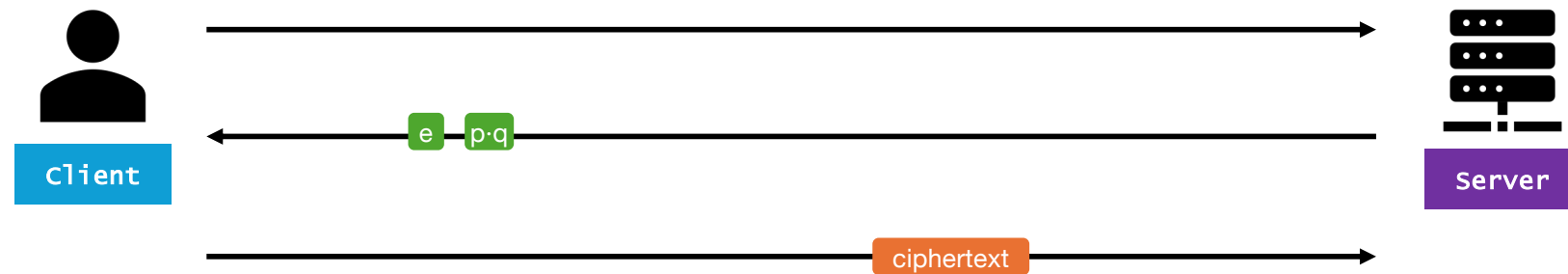
MITM



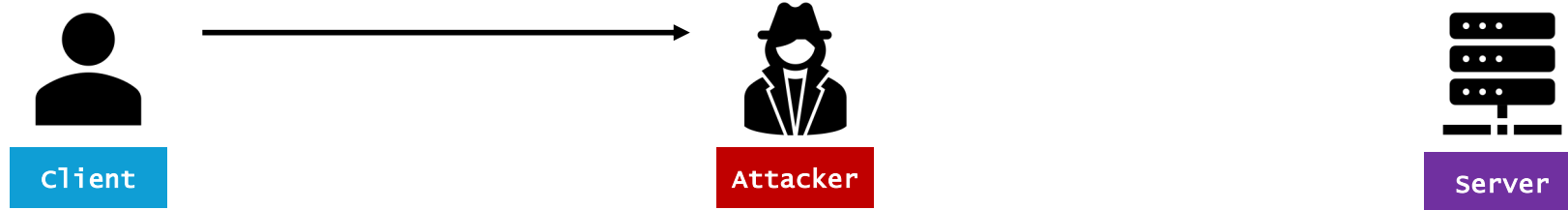
MITM



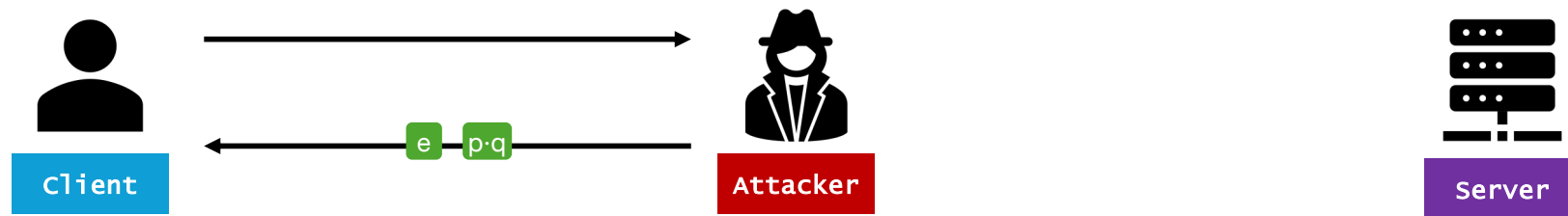
MITM



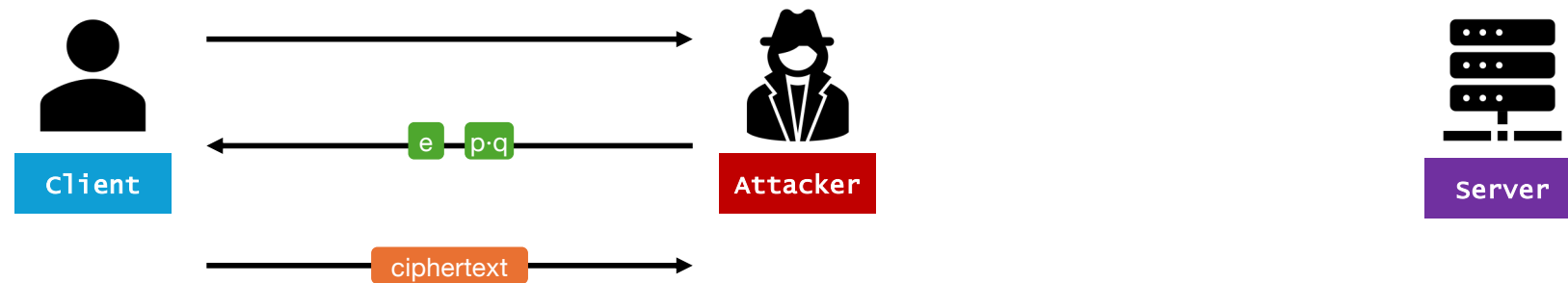
MITM



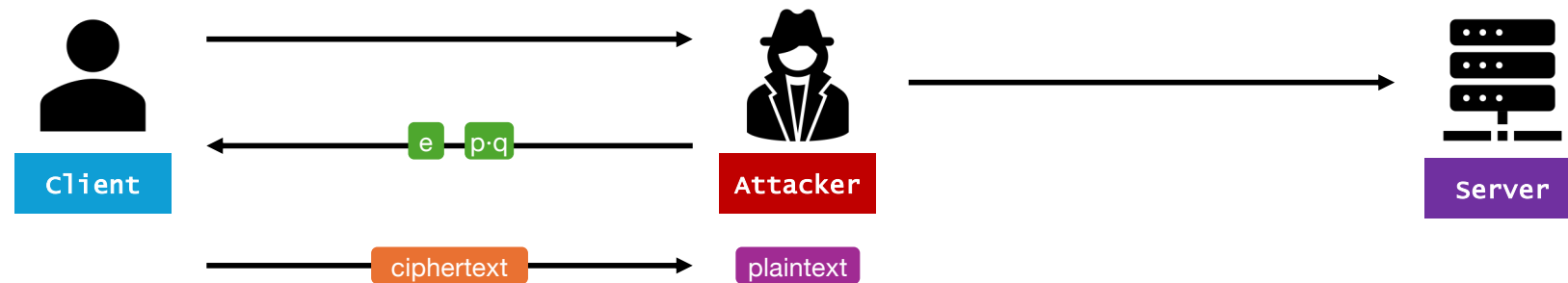
MITM



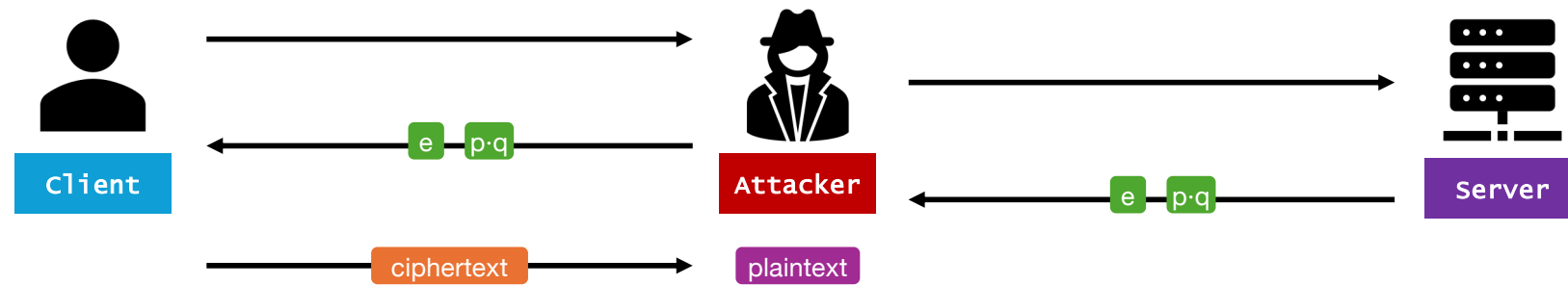
MITM



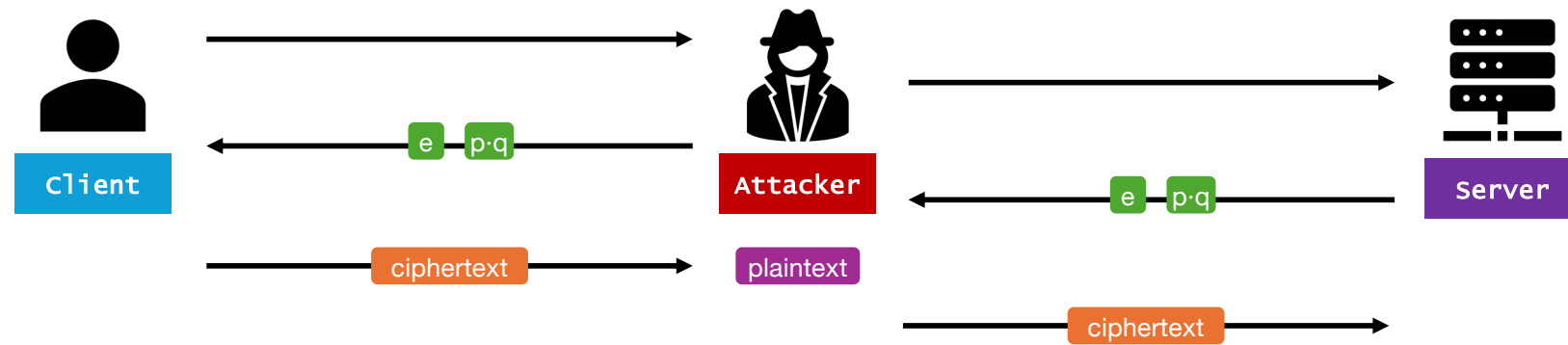
MITM



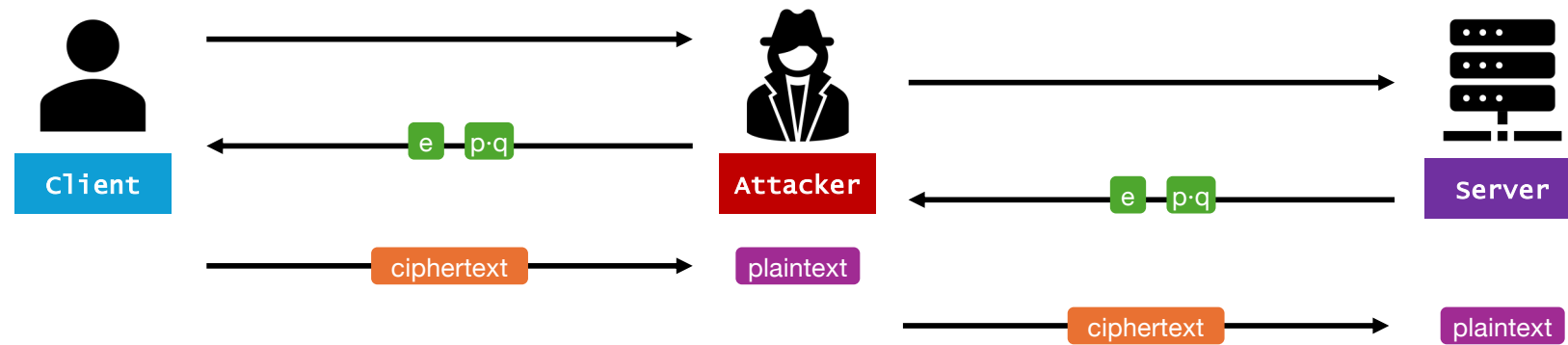
MITM



MITM

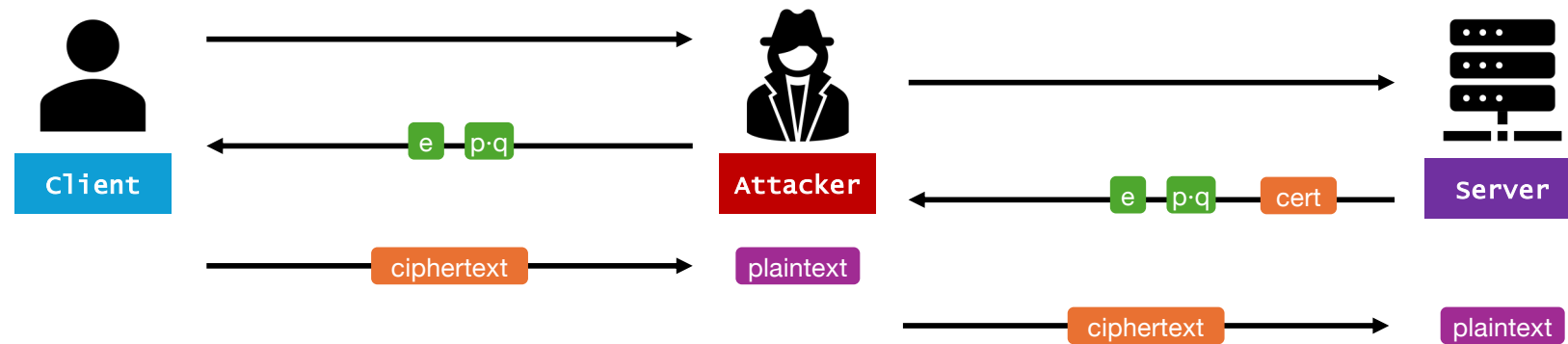


MITM



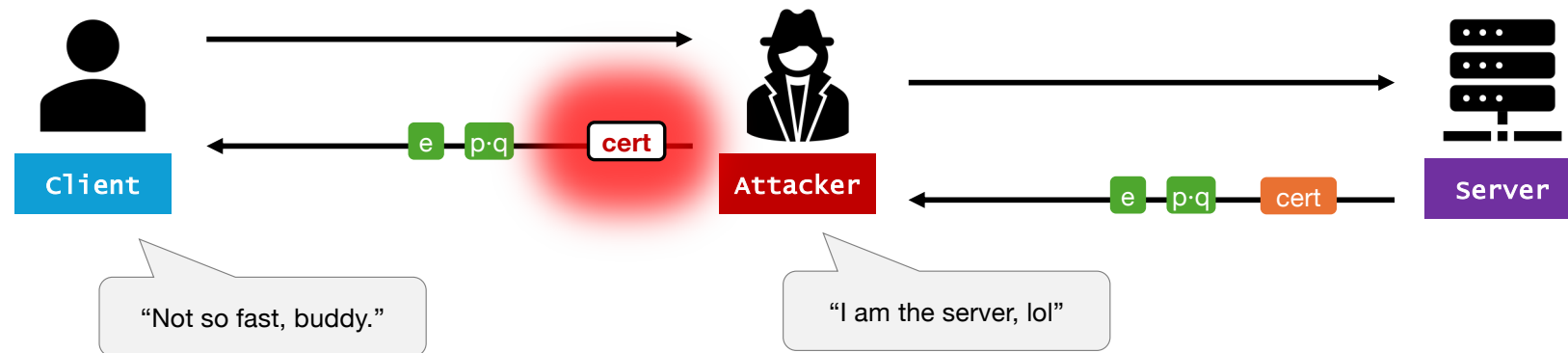
MITM

MITM attacks don't work when the client checks for a certificate:



MITM

MITM attacks don't work when the client checks for a certificate:



MITM

- Your certificate chain is only as strong as its weakest link.
- Be careful with the “trust certificate” checkbox.
- Modern firewalls inspect TLS traffic using a MITM pattern.

Big picture

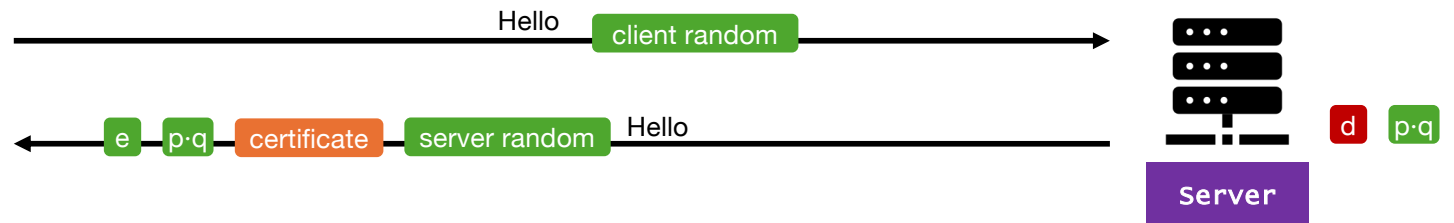
Asymmetric



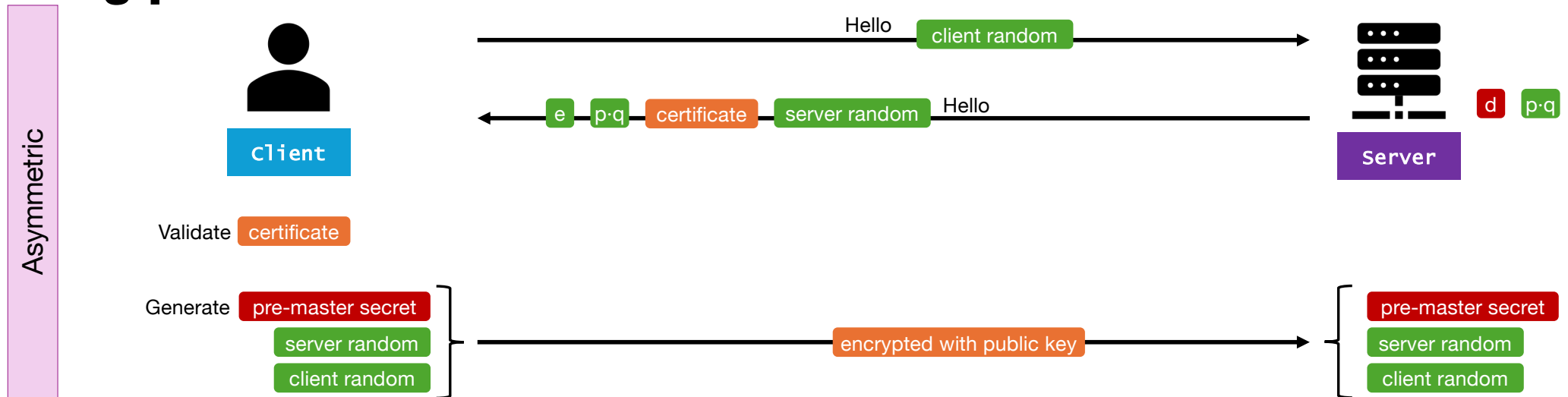
Client

Validate

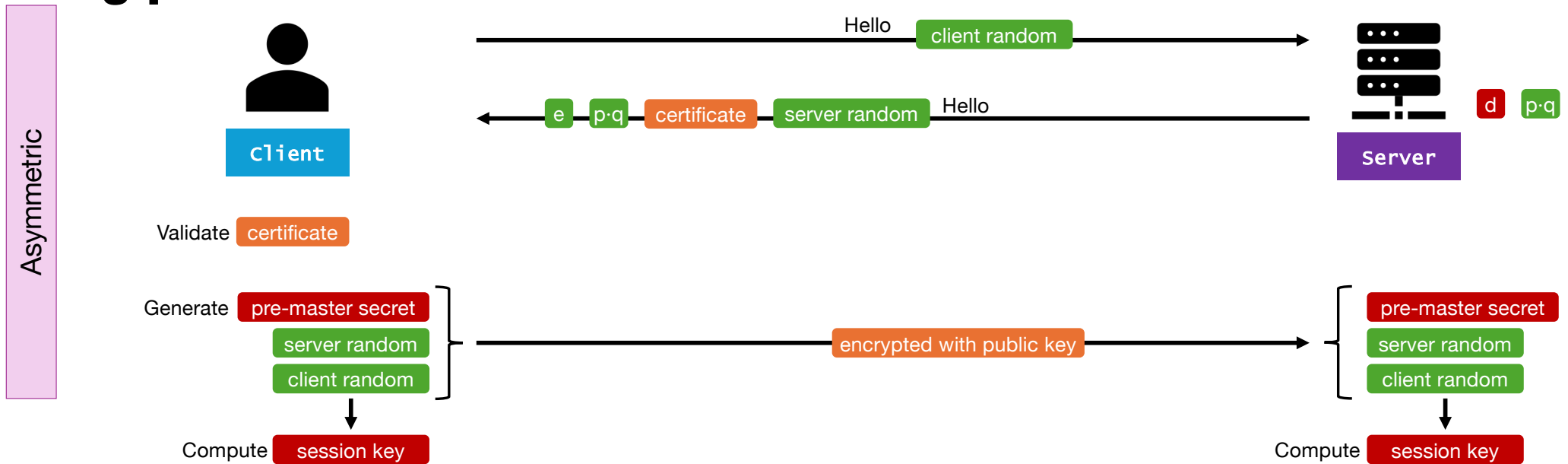
certificate



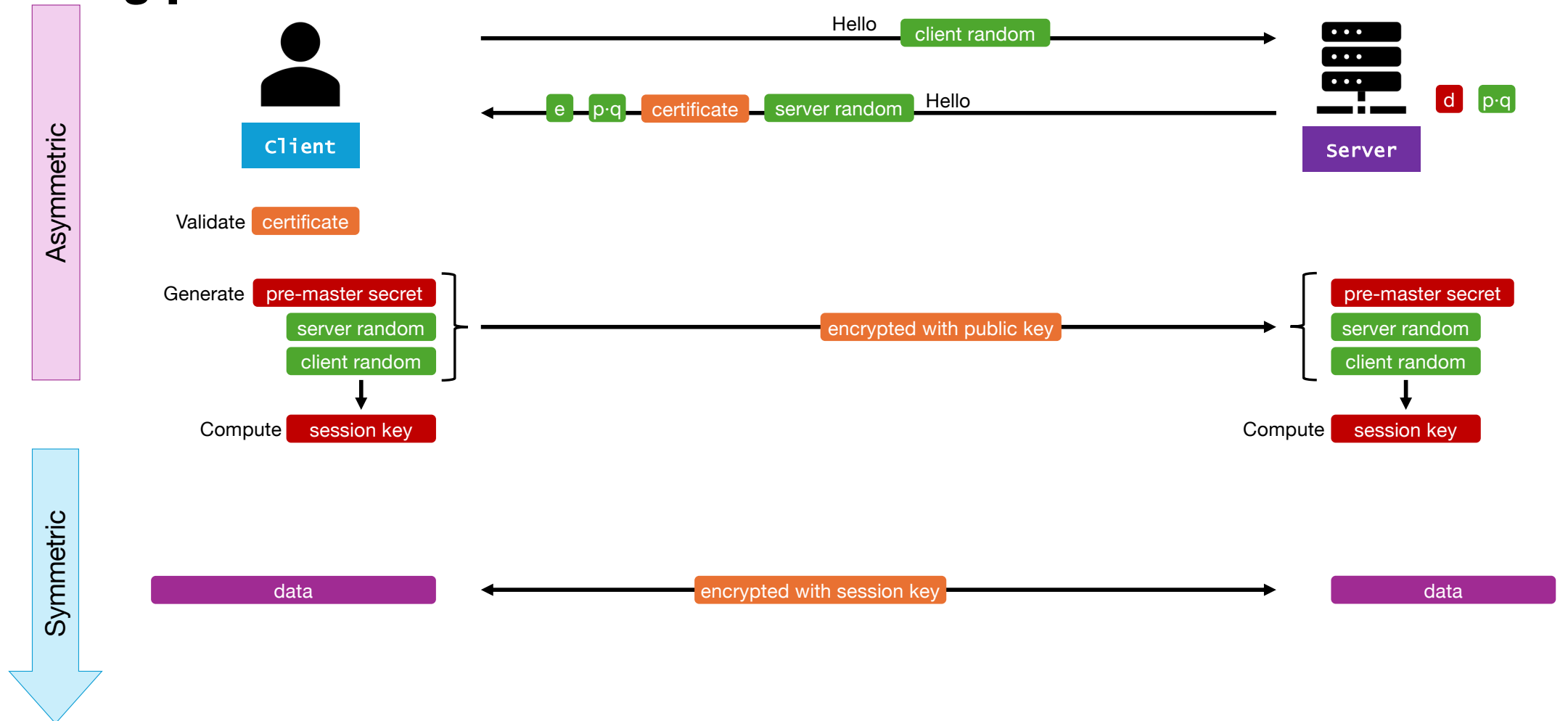
Big picture



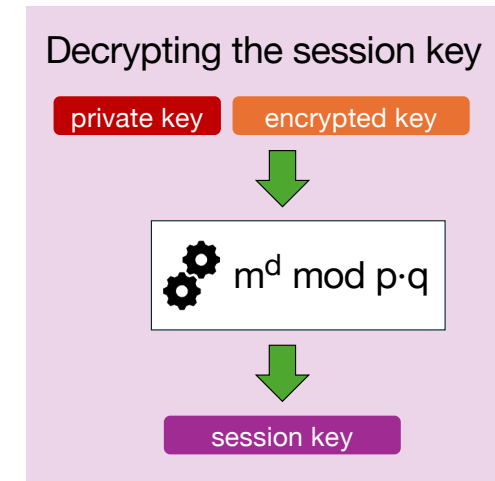
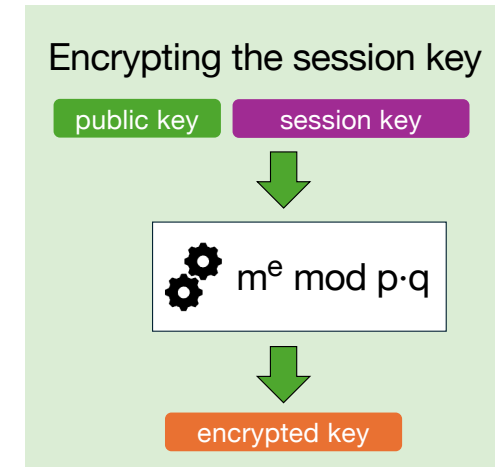
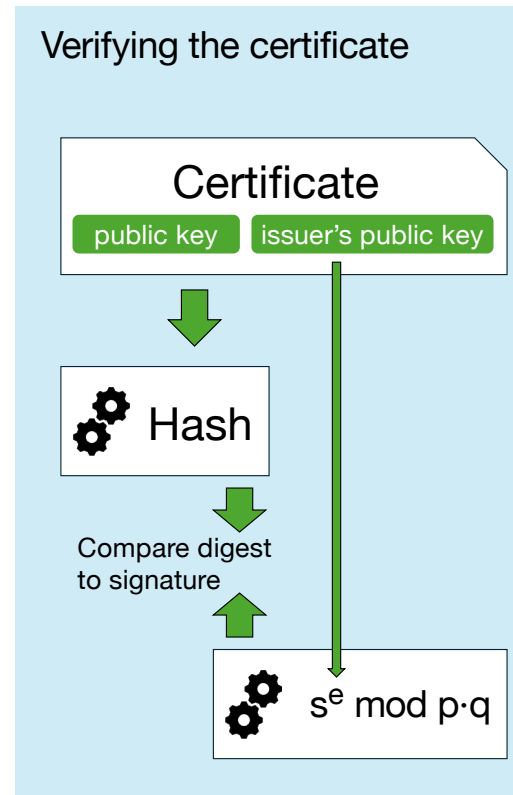
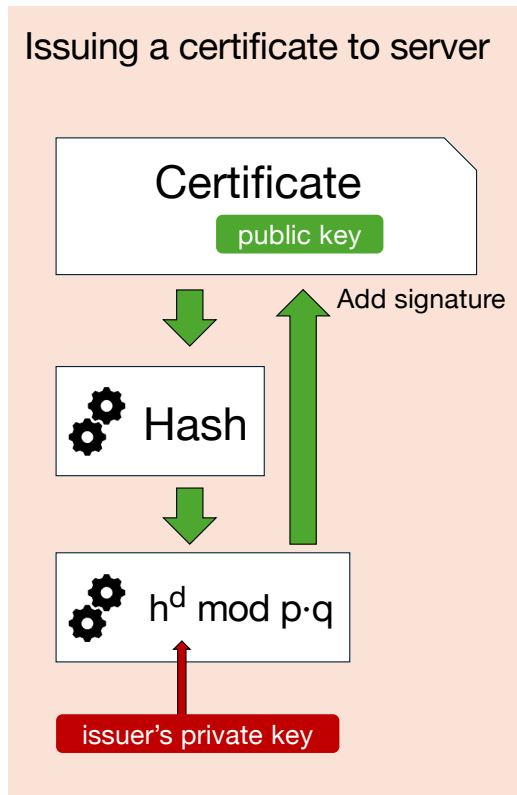
Big picture



Big picture



All the moving parts



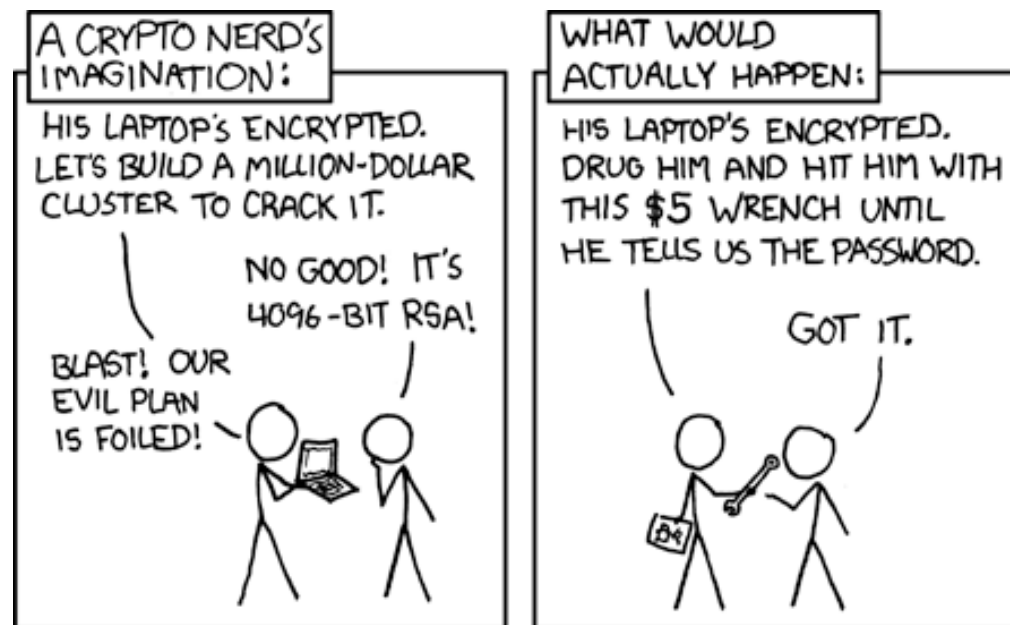
How to defeat strong encryption



How did I beat you?

How to defeat strong encryption

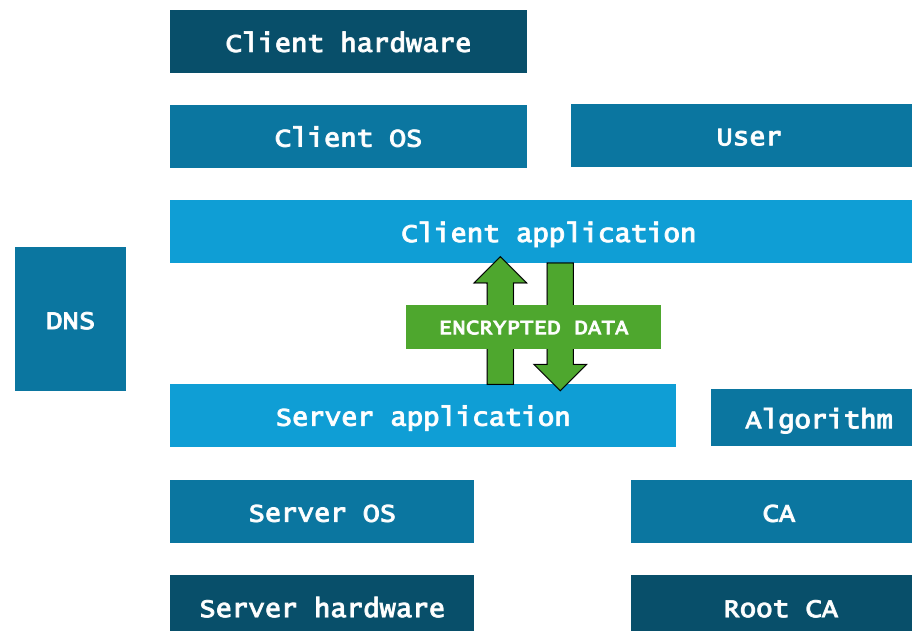
When a cryptographical attack is not feasible, an opponent could try to bypass the encryption entirely.



<https://xkcd.com/538/>

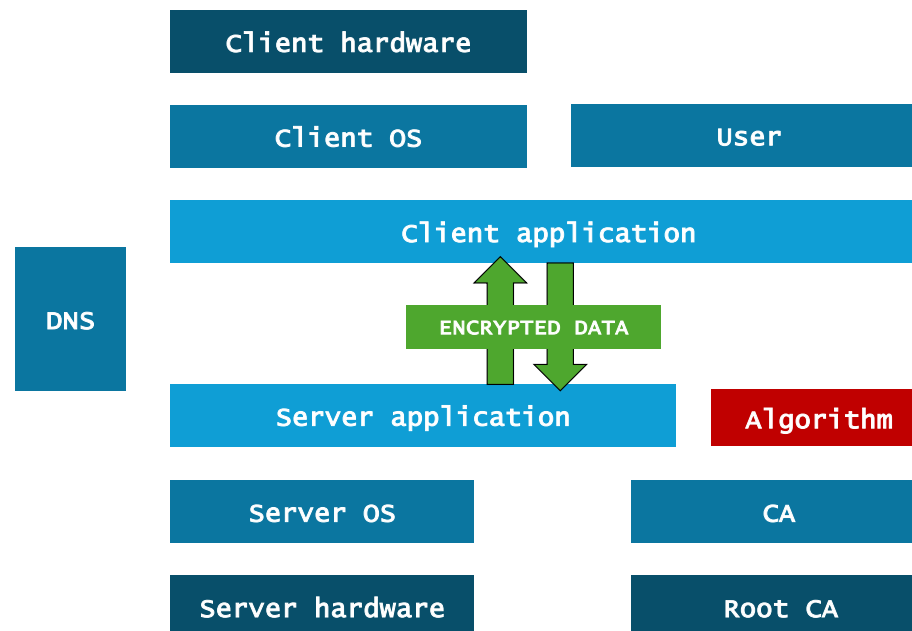
How to defeat strong encryption

When a cryptographical attack is not feasible, an opponent could try to bypass the encryption entirely.



How to defeat strong encryption

When a cryptographical attack is not feasible, an opponent could try to bypass the encryption entirely.



2007: Dual_EC_DRBG backdoor found

Reported by Microsoft, the NSA was shown to have introduced an [intentional backdoor](#) in a random number algorithm used for ECC. Withdrawn in 2014.

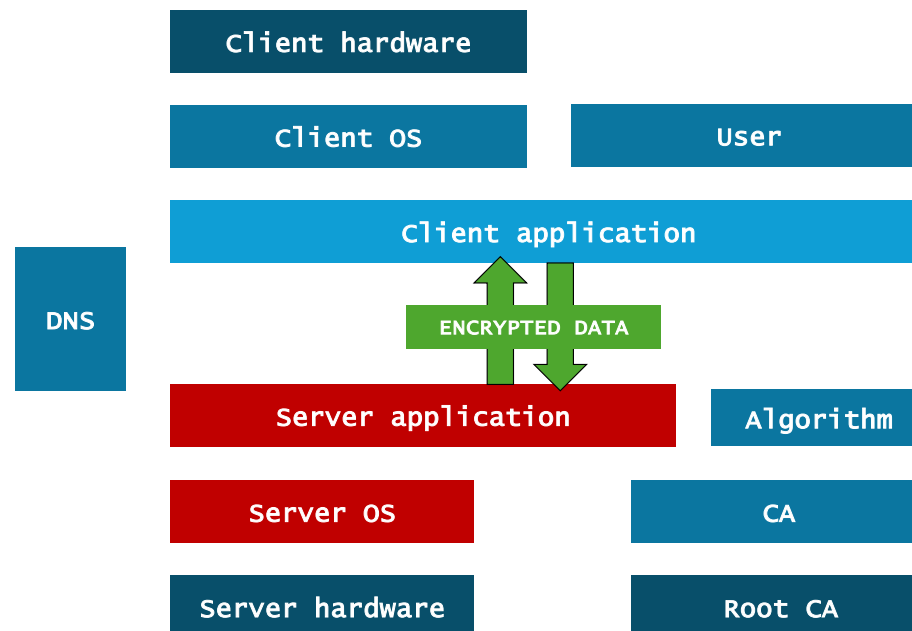
How to defeat strong encryption

When a cryptographical attack is not feasible, an opponent could try to bypass the encryption entirely.

2020: EncroChat

French police installed a “[technical tool](#)” on EncroChat servers, allowing them to covertly log all communications for several months.

See also: [Sky ECC](#)

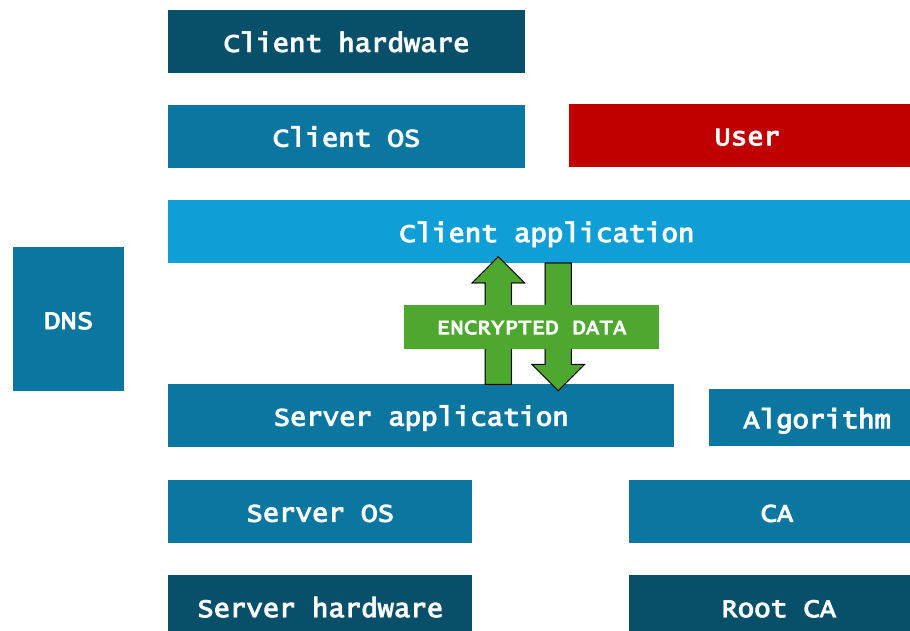


How to defeat strong encryption

When a cryptographical attack is not feasible, an opponent could try to bypass the encryption entirely.

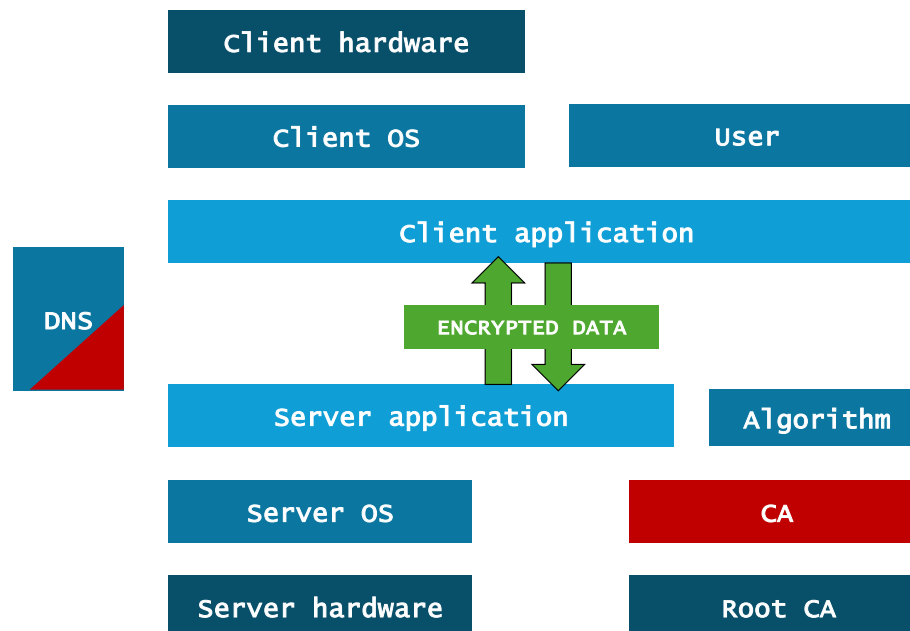
2018-2021: [Trojan Shield](#)

In an elaborate honey pot/trojan attack, law enforcement agencies distributed ANOM, a “secure chat” application, which they then could monitor.



How to defeat strong encryption

When a cryptographical attack is not feasible, an opponent could try to bypass the encryption entirely.



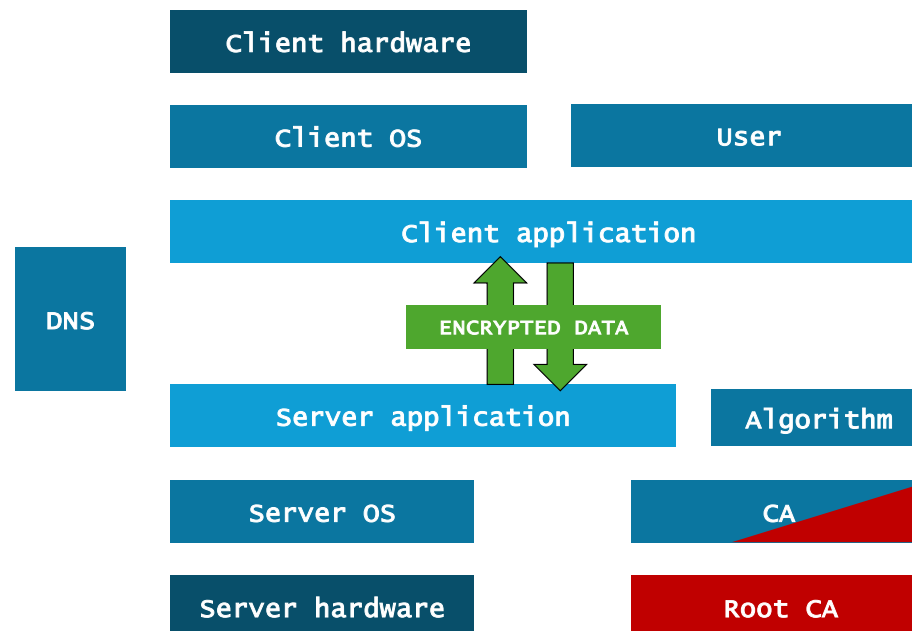
2011: DigiNotar

Dutch CA [DigiNotar](#) was hacked and used to issue fraudulent certificates for Gmail.

The certificates were likely used to spy on 300 000 Iranian users' accounts using a MITM attack.

How to defeat strong encryption

When a cryptographical attack is not feasible, an opponent could try to bypass the encryption entirely.



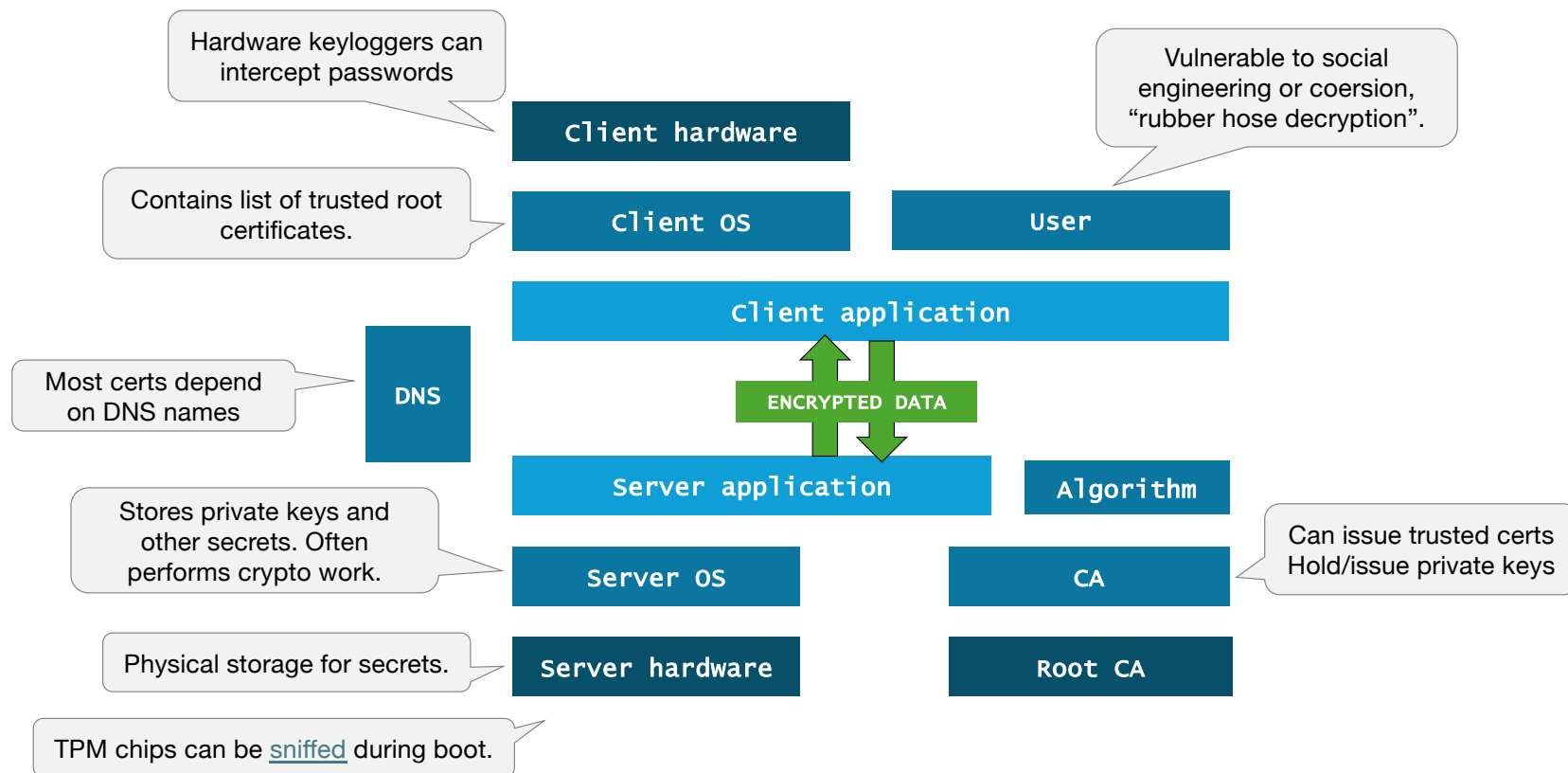
2019: Kazakhstan

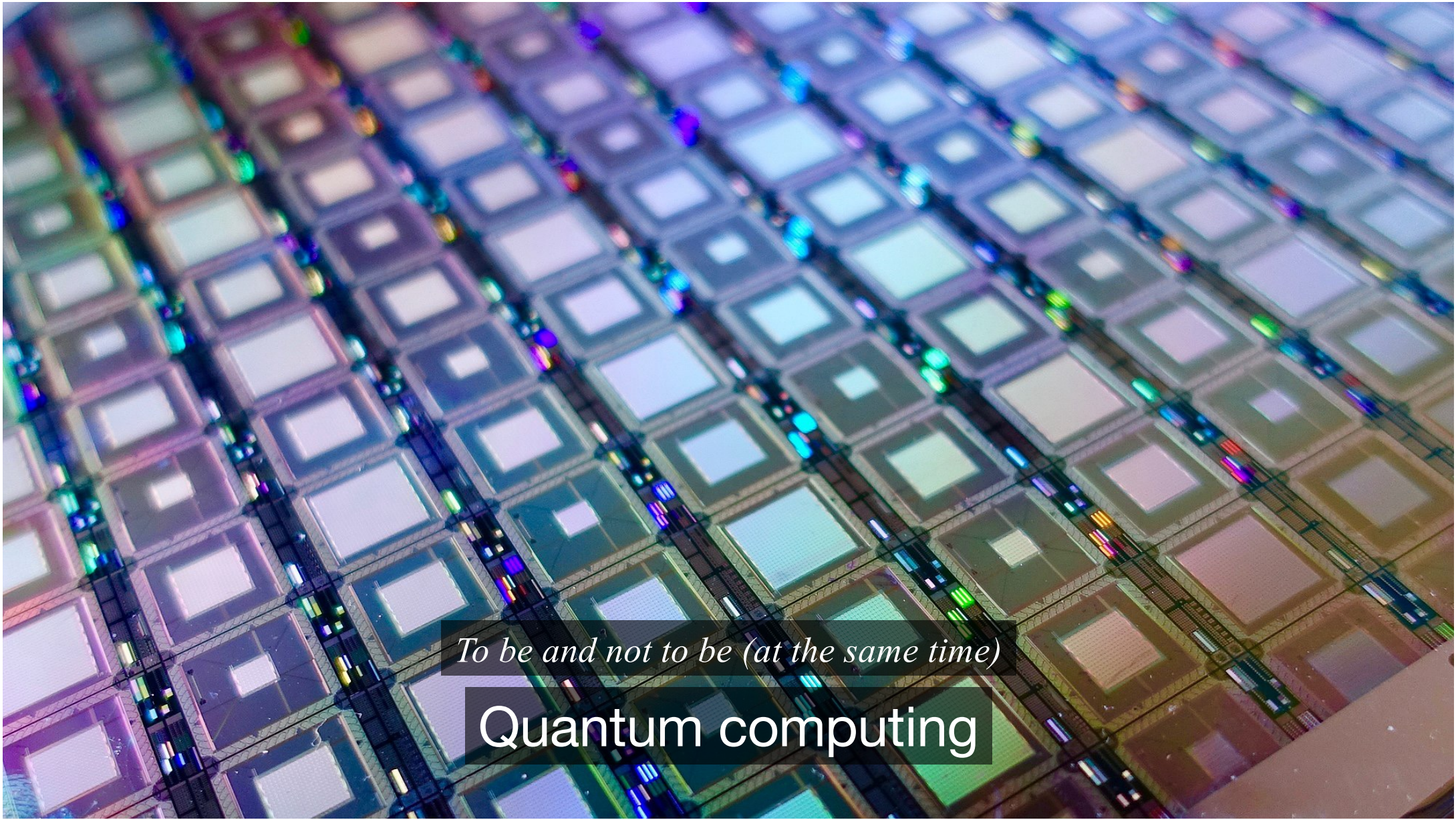
Starting in July 2019, the Kazakh government started instructing citizens to install a root certificate.

Major browser developers have for years refused to add the root cert, and proceeded to block the cert.

How to defeat strong encryption

When a cryptographical attack is not feasible, an opponent could try to bypass the encryption entirely.





To be and not to be (at the same time)

Quantum computing

Quantum computing

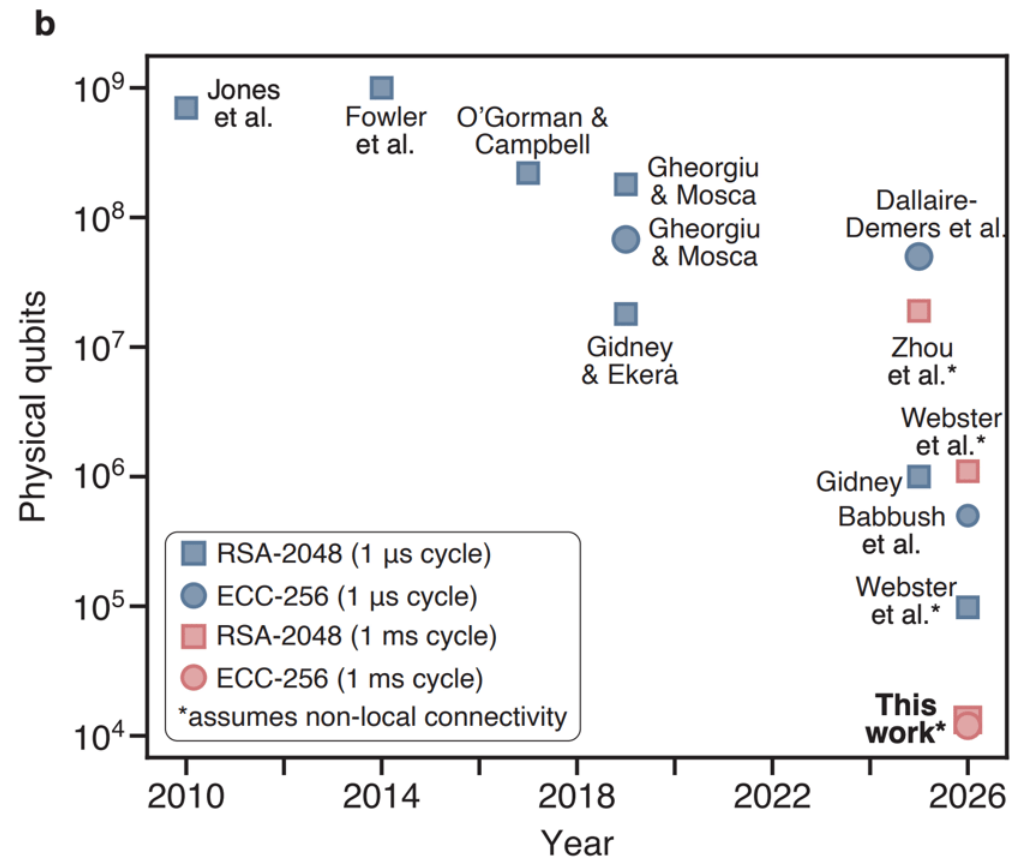
Quantum computers work with "qubits", which are "superpositions" – essentially 0 *and* 1, at the same time.

When you read a qubit, the superposition collapses and returns a regular value based on probability.

A good quantum algorithm aggregates the output values, reinforcing the correct answers and cancelling out the wrong ones, so it becomes *very likely* that you'll get a correct result.



Quantum computing



Source: <https://words.filippo.io/crqc-timeline/> (April 2026)

Quantum computing: Shor's algorithm

$p = 247\,193$

$q = 395\,767$

PRIVATE

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

How do we find the prime factors p and q ?

Quantum computing: Shor's algorithm

$p = 247\,193$

$q = 395\,767$

PRIVATE

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

How do we find the prime factors p and q ?

Select an integer, g , as a reasonable guess for p .

Then find the integer r , such that:

$$g^r \bmod pq = 1$$

Quantum computing: Shor's algorithm

$p = 247\,193$

PRIVATE

$q = 395\,767$

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

$g^r \bmod pq = 1$

$g^r \bmod pq$ is a *periodic* function, meaning that it repeats its output over time.

A quantum algorithm can measure this *period* much, much faster than we could compute it.

r	$g^r \bmod pq$
0	1
...	...
112	9998
113	17625317
114	2306
115	2936784
116	1
117	92299
118	3162
119	1029471
...	...
230	2306
231	2936784
232	1
233	92299
...	...

Quantum computing: Shor's algorithm

$p = 247\,193$

$q = 395\,767$

PRIVATE

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

$g^r \bmod pq = 1$

period of $r=116$.

r	$g^r \bmod pq$
0	1
...	...
112	9998
113	17625317
114	2306
115	2936784
116	1
117	92299
118	3162
119	1029471
...	...
230	2306
231	2936784
232	1
233	92299
...	...

Quantum computing: Shor's algorithm

$p = 247\,193$
 $q = 395\,767$

PRIVATE

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

$g^r \bmod pq = 1$

period of $r=116$.

$g^{116} - 1 = m \cdot pq$

r	$g^r \bmod pq$
0	1
...	...
112	9998
113	17625317
114	2306
115	2936784
116	1
117	92299
118	3162
119	1029471
...	...
230	2306
231	2936784
232	1
233	92299
...	...

Quantum computing: Shor's algorithm

$p = 247\,193$

$q = 395\,767$

PRIVATE

PRIVATE

$p \cdot q = 97\,830\,832\,031$

PUBLIC

$g^r \bmod pq = 1$

period of $r=116$.

$g^{116} - 1 = m \cdot pq$

$(g^{58} + 1)(g^{58} - 1) = m \cdot pq$

r	$g^r \bmod pq$
0	1
...	...
112	9998
113	17625317
114	2306
115	2936784
116	1
117	92299
118	3162
119	1029471
...	...
230	2306
231	2936784
232	1
233	92299
...	...

Quantum computing: Shor's algorithm

$p = 247\ 193$
 $q = 395\ 767$

PRIVATE

PRIVATE

$p \cdot q = 97\ 830\ 832\ 031$

PUBLIC

$g^r \bmod pq = 1$

period of $r=116$.

QUANTUM ALGORITHM

REGULAR ALGORITHM

$$g^{116} - 1 = m \cdot pq$$

$$(g^{58} + 1)(g^{58} - 1) = m \cdot pq$$

$$\gcd(g^{58} + 1, pq) = p$$

r	$g^r \bmod pq$
0	1
...	...
112	9998
113	17625317
114	2306
115	2936784
116	1
117	92299
118	3162
119	1029471
...	...
230	2306
231	2936784
232	1
233	92299
...	...

Mmmkay

Please take a minute to give me
some feedback:



Mmmkay

github.com/sqlsunday/presentations

Feel free to reach out with questions
or comments.

Email: daniel@strd.co
Blog: sqlsunday.com
Bluesky: [@dhma.ch](https://dhma.ch)



Tools, links, docs

AES:

- AES visualizer: http://advancedcomputing.org/aes_visualizer.html

RSA:

- Interactive RSA calculation: <https://www.henryschmale.org/2022/03/14/rsa.html>
- RSA numbers: https://en.wikipedia.org/wiki/RSA_numbers
- Big number calculator: <https://defuse.ca/big-number-calculator.htm>
- Prime factorization tool:
<http://www.javascripter.net/math/calculators/primefactorscalculator.htm>
- Fermat attack on RSA (when p and q are close to the square root):
<https://fermatattack.secvuln.info/>

Quantum:

- A Cryptography Engineer's Perspective on Quantum Computing Timelines:
<https://words.filippo.io/crqc-timeline/>
- Veritasium on Shor's Algorithm (beginner-friendly)
<https://youtu.be/-UrdExQW0cs>
- 3Blue1Brown on Grover's Algorithm (advanced)
<https://youtu.be/RQWpF2Gb-gU>